Dark Energy as Residual Vacuum Free Energy: A Thermodynamic Bound on the Cosmological Constant

Part of the FST Programme [35]

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Abstract

The cosmological constant problem remains one of the deepest puzzles in theoretical physics: naive quantum field theory estimates of the vacuum energy density exceed the observed value by roughly 120 orders of magnitude. We propose a thermodynamic variational principle that constrains the effective cosmological constant Λ_{eff} without introducing a fitted late-time Λ parameter; the quantitative scaling is conditional on the RG-matching assumptions stated below. Starting from a free-energy function $\Phi(\rho_{\text{vac}})$ defined over trial vacuum energy densities and regularised between an ultraviolet cutoff Λ_{UV} and an infrared cutoff set by the Hubble radius H_0^{-1}, we argue that, under this matching, the unique minimum of Φ yields a residual vacuum energy density scaling as $\Lambda_{\text{eff}} \sim H_0^2/\ln(M_{\text{Pl}}/H_0)$. The resulting numerical estimate lies within an order of magnitude of the Planck 2018 value. Embedding the variational bound in a scalar-tensor Lagrangian ($R+\gamma R^2$ plus a Pöschl–Teller scalar field), we derive a tanh saturation dynamics for the dark-energy density parameter, an effective equation of state in the thawing regime, and a falsifiable target for the gravitational slip parameter $\eta = 5/4$ in the quasi-static, linear, unscreened sub-Compton regime of metric $f(R)$ theories, including the Hu–Sawicki class used for chameleon screening (see Remark 5.14). The framework naturally addresses the coincidence problem by linking the vacuum energy to the current expansion rate.

Limitations (honest scope statement). The scaling $\Lambda_{\text{eff}} \sim H_0^2/\ln(M_{\text{Pl}}/H_0)$ follows from the variational principle *conditional* on a renormalisation-group matching that contributes a second logarithm $\ln(M_{\text{Pl}}/H_0)$ from integrating $\beta_\Lambda^{\text{QQG}}$ between H_0 and M_{Pl}. This RG-matching is not derived from first principles in the present paper. A structured reduction onto two explicit open sub-conditions ((B1) existence of a perturbatively accessible QQG renormalisation-group trajectory ending in a Hu–Sawicki-type effective $f(R)$; (B2) the $\beta_\Lambda^{\text{QQG}}$-integral evaluating to $\ln(M_{\text{Pl}}/H_0)$ with controlled $O(1)$ prefactor) is documented in the companion proof note on the bridge to Hu–Sawicki screening. We classify this as a Cosmology-Pattern-A reduction (sub-reduction with unsolved root): the present paper structures the cosmological constant problem rather than closing it.

Keywords: cosmological constant, dark energy, vacuum energy, scalar-tensor gravity, gravitational slip

1 Introduction

The cosmological constant Λ was introduced by Einstein in 1917 to allow a static universe and later abandoned after the discovery of cosmic expansion. Its modern reincarnation as dark energy – the agent driving the accelerated expansion observed since 1998 [6, 7] – has brought it back to the centre of fundamental physics.

1.1 The cosmological constant problem

In quantum field theory every field contributes zero-point energy to the vacuum. A straightforward calculation with a Planck-scale cutoff gives [1, 2]

$$\rho_{\text{vac}}^{\text{QFT}} \sim \frac{M_{\text{Pl}}^4}{16\pi^2} \approx 10^{74}\ \text{GeV}^4, \tag{1}$$

while the observed dark-energy density is

$$\rho_{\text{vac}}^{\text{obs}} \approx 10^{-47}\ \text{GeV}^4, \tag{2}$$

a discrepancy of approximately 10^{120}.

1.2 Existing approaches

Several strategies have been pursued to resolve or circumvent this discrepancy:

- **Anthropic selection** in a landscape of vacua [26, 27].
- **Quintessence** – a slowly rolling scalar field whose potential energy mimics Λ [3].
- **Modified gravity** – replacing General Relativity at cosmological scales ($f(R)$ theories, massive gravity).
- **Sequestering mechanisms** that decouple the vacuum energy from curvature [28].

Each approach resolves some aspects but introduces new free parameters or conceptual difficulties.

1.3 Our approach

We take a different route. Rather than cancelling the vacuum energy or invoking selection effects, we argue that the *observed* cosmological constant is the thermodynamic equilibrium value of a free-energy functional defined over vacuum fluctuations. The functional is bounded below by the infrared geometry of spacetime, and its unique minimum naturally produces a residual energy density of order H_0^2/G.

This variational architecture realises the **"Pattern B" (Flow Selection)** logic of the FST framework, as pioneered in the **gauge-theoretic reformulation of the Riemann Hypothesis** (see [19]). Just as the stability of the arithmetical kernel is ensured by a second-order resolvent dominance that identifies the "physical" zeros, the cosmological constant emerges here as the "selection gauge" that stabilises the vacuum against runaway fluctuations. The functional $\Phi(\rho)$ is the physical analogue of the renormalised Li functional in the RH case: both identify a unique, non-negative minimiser as the attractor of the complex system. Dark energy is thus the "selection pressure" required for gravitational stability.

This paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST programme are cited for context but are not logically required.

2 The Vacuum Free-Energy Functional

2.1 Setup and definitions

Consider a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) background with scale factor $a(t)$ and Hubble parameter $H = \dot{a}/a$. We define a free-energy function over trial vacuum energy densities $\rho > 0$:

Definition 2.1 (Vacuum free-energy function). Let Λ_{UV} and Λ_{IR} denote ultraviolet and infrared momentum cutoffs respectively. The *vacuum free-energy function* is

$$\Phi(\rho) \;=\; \int_{\Lambda_{\mathrm{IR}}}^{\Lambda_{\mathrm{UV}}} \left[\frac{1}{2}\,\omega(k) \;+\; T_{\mathrm{dS}}\,\rho\,\ln\!\left(\frac{\rho}{\rho_{\mathrm{Pl}}}\right) \;+\; V_{\mathrm{grav}}(\rho,\,k) \right] \frac{4\pi\,k^2}{(2\pi)^3}\,\mathrm{d}k, \tag{3}$$

where $\rho > 0$ is a trial vacuum energy density (a scalar parameter, not a field) and:

- $\omega(k) = \sqrt{k^2 + m^2}$ is the mode frequency (summed over all species),
- $T_{\mathrm{dS}} = H/(2\pi)$ is the de Sitter (Gibbons–Hawking) temperature associated with the cosmological horizon,
- $\rho_{\mathrm{Pl}} = M_{\mathrm{Pl}}^4$ is the Planck energy density,
- $V_{\mathrm{grav}}(\rho, k)$ encodes the gravitational back-reaction of the vacuum energy on the mode structure.

The entropy term $\rho\,\ln(\rho/\rho_{\mathrm{Pl}})$ has the standard thermodynamic form $F = E - TS$ with the Boltzmann entropy density $s(\rho) = -\rho\,\ln(\rho/\rho_{\mathrm{Pl}})$.

Remark 2.2. The entropic term $T_{\mathrm{dS}}\,\rho\,\ln(\rho/\rho_{\mathrm{Pl}})$ acts as a thermodynamic penalty. For $\rho \to 0^+$, the product $\rho\,\ln(\rho/\rho_{\mathrm{Pl}}) \to 0^-$, so the entropy contribution vanishes; the gravitational back-reaction term $V_{\mathrm{grav}} \propto \rho^2 \to 0$ likewise vanishes, and Φ reduces to the (finite, ρ-independent) zero-point energy. For $\rho \to +\infty$, both $\rho\ln\rho$ and ρ^2 diverge, driving $\Phi \to +\infty$. The minimum therefore occurs at an intermediate value $\rho_* > 0$.

Remark 2.3 (Dimensional convention). In natural units the three terms under the integral in (3) carry different engineering dimensions: $[\omega(k)] = [\text{energy}]$, $[T_{\mathrm{dS}}\,\rho\,\ln(\rho/\rho_{\mathrm{Pl}})] = [\text{energy}]^5$, and $[G\rho^2/k^2] = [\text{energy}]^4$. This reflects the *effective* character of the functional: each term is implicitly accompanied by dimensionless coupling constants of order unity (absorbed into the coefficients B and C in the proof of Theorem 3.1). The physically meaningful content of the variational principle resides in the ρ-dependence of each term, which determines the location of the minimum. The absolute magnitude of Φ is not observable; only the stationarity condition $\mathrm{d}\Phi/\mathrm{d}\rho = 0$ has physical content, and this condition involves only ratios of the coefficients B/C, where the implicit dimensional factors cancel. A fully dimensionally homogeneous formulation—e.g., recasting Φ in terms of the dimensionless ratio ρ/ρ_{Pl}—is deferred to the full version. **Warning:** Until this reformulation is carried out, the implicit dimensional factors are not merely "of order unity" but necessarily involve powers of M_{Pl} and/or H_0—precisely the scales whose ratio determines the result. The claim that the stationarity condition $\mathrm{d}\Phi/\mathrm{d}\rho = 0$ involves only the ratio B/C is correct, but the physical content of B/C depends on these implicit factors, which must be made explicit to verify the scaling law.

Remark 2.4 (Dimensionless formulation). All terms in the functional (3) are rendered dimensionless by measuring densities in units of $\rho_{\mathrm{Pl}} = c^5/(\hbar G^2)$, wavenumbers in units of the Planck length ℓ_{Pl}, and energies in units of $M_{\mathrm{Pl}}c^2$. In these natural units, the entropy coefficient T_{dS}, the gravitational coupling G, and the mode energy $\omega(k)$ all carry $O(1)$ dimensionless prefactors. The coupling constants alluded to in the text are precisely these Planck-unit normalisations.

2.2 Choice of cutoffs

The ultraviolet cutoff is set at the reduced Planck scale:

$$\Lambda_{\mathrm{UV}} \;=\; M_{\mathrm{Pl}} \;=\; \left(\frac{\hbar c}{8\pi G}\right)^{1/2} \;\approx\; 2.4 \times 10^{18}\ \mathrm{GeV}. \tag{4}$$

The infrared cutoff is identified with the Hubble radius:

$$\Lambda_{\mathrm{IR}} \;=\; H_0 \;\approx\; 1.5 \times 10^{-42}\ \mathrm{GeV}. \tag{5}$$

This identification has precedent in holographic dark-energy models [25] and in the UV-IR cutoff relation of Cohen, Kaplan, and Nelson [34], who argue that in an effective field theory coupled to gravity the IR and UV cutoffs are related by $\Lambda_{\mathrm{UV}}^2\, L \lesssim M_{\mathrm{Pl}}$, with $L \sim H_0^{-1}$ the cosmological horizon size. The identification is physically motivated by the causal structure of de Sitter space.

2.3 Gravitational back-reaction term

We model V_{grav} as the leading-order self-gravitational energy cost of sustaining a vacuum energy density ρ at scale k:

$$V_{\mathrm{grav}}(\rho,\, k) \;=\; \frac{8\pi G\,\rho^2}{k^2}. \tag{6}$$

This quadratic dependence on ρ ensures that $\Phi(\rho)$ is strictly convex, which is essential for the uniqueness of the minimum.

Lemma 2.5 (Minimality of the back-reaction form)**.** *Let $V[\rho, k]$ be a gravitational back-reaction potential satisfying:*

(M1) ***Gravitational coupling:*** *V is proportional to exactly one power of Newton's constant G.*

(M2) ***Dimensional consistency:*** *In natural units ($c = \hbar = 1$), $[V] = [\mathrm{energy}]^4$ (energy density per mode).*

(M3) ***Minimal nonlinearity:*** *V is the* lowest-order *monomial in ρ consistent with self-interaction (i.e., at least quadratic in ρ).*

(M4) ***IR dominance:*** *V grows as k decreases (infrared-dominated back-reaction).*

Then, up to a dimensionless constant $c_0 > 0$,

$$V[\rho, k] \;=\; c_0\,\frac{G\,\rho^2}{k^2}.$$

The conventional choice $c_0 = 8\pi$ (incorporating mode-counting factors) is adopted in (6).

Proof. Write $V = G^\alpha \rho^\beta k^\gamma$ with α, β, γ to be determined. In natural units, $[G] = [\text{energy}]^{-2}$, $[\rho] = [\text{energy}]^4$, $[k] = [\text{energy}]^1$. The dimensional constraint $[V] = [\text{energy}]^4$ requires

$$-2\alpha + 4\beta + \gamma = 4.$$

By (M1), $\alpha = 1$, giving $\gamma = 6 - 4\beta$. By (M3), $\beta \geq 2$. The minimal choice $\beta = 2$ yields $\gamma = -2$, hence $V \sim G\,\rho^2/k^2$. The next-order term ($\beta = 3$, $\gamma = -6$) gives $V \sim G\,\rho^3/k^6$, which is suppressed by a factor ρ/k^4 relative to the leading term—negligible in the regime $\rho \ll k^4$ (i.e., below the Planck density). Finally, (M4) requires $\gamma < 0$, which is satisfied by $\gamma = -2$.

The value of c_0 is determined by matching to the gravitational self-energy. The Newtonian potential energy of a uniform sphere of density ρ and radius $R = k^{-1}$ is $U = -\frac{16\pi^2}{15}\,G\,\rho^2\,R^5$, giving a self-energy density $u = |U|/(4\pi R^3/3) = \frac{4\pi}{5}\,G\,\rho^2\,R^2 = \frac{4\pi}{5}\,G\,\rho^2/k^2$. The 8π coefficient in (6) represents a conventional normalisation that incorporates the mode-counting factor from the k-space integration measure $(4\pi k^2/(2\pi)^3)$. The dimensionless constant c_0 is thus of order $O(4\pi)$–$O(8\pi)$; its precise value does not affect the parametric scaling of the minimum and can be absorbed into the renormalisation of Λ_{ren}. □

Remark 2.6. Lemma 2.5 addresses the principal criticism that V_{grav} is an *ansatz* rather than a first-principles derivation. While the lemma does not derive V_{grav} from the Einstein equations, it shows that the functional form (6) is *uniquely determined* (up to a numerical coefficient) by four physically transparent axioms. Any alternative back-reaction term would either violate dimensional consistency, introduce higher-order gravitational coupling (G^2 terms, suppressed by M_{Pl}^{-2}), or be sub-leading in ρ.

3 Main Result

Theorem 3.1 (Thermodynamic bound on Λ_{eff} – conditional)**.** *Under the assumptions of Definition 2.1, the function $\Phi(\rho)$ attains a unique global minimum at $\rho = \rho_* > 0$; this part (existence, uniqueness, strict convexity) is proven. The corresponding effective cosmological constant*

$$\Lambda_{\text{eff}} \;=\; \frac{8\pi G}{c^4}\,\rho_* \tag{7}$$

satisfies the conjectured *bound*

$$\Lambda_{\text{eff}} \;\sim\; \frac{H_0^2}{c^2}\,\left[\ln\!\left(\frac{M_{\text{Pl}}}{H_0}\right)\right]^{-1}. \tag{8}$$

*The scaling (8) does **not** follow from the raw mode sum alone; the raw stationarity condition yields $\rho_* \sim H_0 M_{\text{Pl}}^4 \ln(M_{\text{Pl}}/H_0)$, which is $\sim 10^{60}$ times too large. The physically* *relevant scaling requires the renormalisation-group matching of Section 5.3, where the UV-divergent contributions are absorbed into counterterms and only the infrared running survives. This matching is* externally imposed*, not derived from the variational principle itself; its rigorous derivation from the functional Φ remains the principal open problem (see "Proof status summary" below).*

Proof (sketch). Write $\Phi(\rho) = A + B\, f(\rho) + C\,\rho^2$, where $A = \int_{\Lambda_{\rm IR}}^{\Lambda_{\rm UV}} \frac{1}{2}\omega(k)\,\mathrm{d}\mu(k)$ is the (ρ-independent) zero-point energy, $B = T_{\rm dS}\int_{\Lambda_{\rm IR}}^{\Lambda_{\rm UV}} \mathrm{d}\mu(k) > 0$ with $\mathrm{d}\mu(k) = 4\pi k^2/(2\pi)^3\,\mathrm{d}k$, $f(\rho) = \rho\,\ln(\rho/\rho_{\rm Pl})$, and $C = 8\pi G\int_{\Lambda_{\rm IR}}^{\Lambda_{\rm UV}} k^{-2}\,\mathrm{d}\mu(k) > 0$.

Step 1: Existence. As $\rho \to 0^+$: $f(\rho) = \rho\ln(\rho/\rho_{\rm Pl}) \to 0$ and $C\rho^2 \to 0$, so $\Phi(\rho) \to A$ (finite, from below, since $f(\rho) < 0$ for $\rho < \rho_{\rm Pl}$). As $\rho \to +\infty$: $C\rho^2$ dominates and $\Phi \to +\infty$. Since Φ is continuous on $(0,\infty)$, tends to A as $\rho \to 0^+$, and to $+\infty$ as $\rho \to +\infty$, it attains its infimum at some $\rho_* > 0$ (the infimum is finite and strictly less than A, hence it is a minimum).

Step 2: Uniqueness. The second derivative is

$$\frac{\mathrm{d}^2\Phi}{\mathrm{d}\rho^2} \;=\; \int_{\Lambda_{\rm IR}}^{\Lambda_{\rm UV}} \left[\frac{T_{\rm dS}}{\rho} \;+\; \frac{16\pi G}{k^2}\right]\frac{4\pi k^2}{(2\pi)^3}\,\mathrm{d}k \;=\; \frac{B}{\rho} + 2C \;>\; 0 \quad \text{for all } \rho > 0, \tag{9}$$

since $B, C > 0$. Strict convexity implies the minimum is unique.

Step 3: Scaling. Setting $\mathrm{d}\Phi/\mathrm{d}\rho = 0$ yields the stationarity condition

$$B\Big[1 + \ln(\rho_*/\rho_{\rm Pl})\Big] \;+\; 2C\,\rho_* \;=\; 0. \tag{10}$$

Since $\rho_* \ll \rho_{\rm Pl}$, the logarithm $\ln(\rho_*/\rho_{\rm Pl})$ is large and negative, so the first term is dominated by $-B\,\ln(\rho_{\rm Pl}/\rho_*)$ and the balance reads

$$\rho_* \;\approx\; \frac{B}{2C}\left[\ln\!\left(\frac{\rho_{\rm Pl}}{\rho_*}\right) - 1\right]. \tag{11}$$

A direct evaluation of the mode integrals B and C involves the ratio I_1/I_2 where $I_1 = \int \mathrm{d}\mu(k) \sim \Lambda_{\rm UV}^3$ and $I_2 = \int k^{-2}\,\mathrm{d}\mu(k) \sim \Lambda_{\rm UV}$, so that $B/C \sim T_{\rm dS}\,\Lambda_{\rm UV}^2/(8\pi G) = H_0\,M_{\rm Pl}^2/(16\pi^2\,G)$. In natural units ($G = M_{\rm Pl}^{-2}$), this gives $B/C \sim H_0\,M_{\rm Pl}^4/(16\pi^2)$.

The raw mode-sum scaling yields $\rho_* \sim H_0\,M_{\rm Pl}^4\,\ln(M_{\rm Pl}/H_0)$, which is far too large. This reflects the well-known UV sensitivity of vacuum-energy calculations: the mode sum is dominated by UV modes and does not directly give the observed value. As discussed in Section 5.3, the physically meaningful statement is obtained after renormalisation-group matching, where the UV-divergent pieces are absorbed into counterterms and only the *infrared running* survives. In the renormalised framework, the logarithmic factor $\ln(M_{\rm Pl}/H)$ arises from running between $\mu = M_{\rm Pl}$ and $\mu = H_0$, yielding

$$\rho_*^{\rm ren} \;\sim\; \frac{H_0^2}{8\pi G}\,\frac{1}{\ln(M_{\rm Pl}/H_0)}, \tag{12}$$

which gives (8) after inserting into (7). The derivation of this scaling from renormalised quantities is presented in Section 5.3; the sketch above establishes only the structural properties (existence, uniqueness, competition between entropic and gravitational terms).

A rigorous proof requires control of the mode sum over all Standard Model species and a proper renormalisation of the UV divergences. In particular, Step 3 above establishes only *the structural competition between entropic and gravitational terms; the physical scaling $\rho_*^{\rm ren} \sim H_0^2/(8\pi G \ln(M_{\rm Pl}/H_0))$ is obtained only after the renormalisation-group matching of Section 5.3 and does not follow from the raw mode-sum functional (3) alone. The full derivation will be addressed in the complete version.* □

Corollary 3.2 (Numerical estimate)**.** *With $H_0 \approx 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \approx 2.2 \times 10^{-18}\ \mathrm{s^{-1}}$ and $\ln(M_{\rm Pl}/H_0) \approx 140$, we obtain*

$$\Lambda_{\rm eff} \ \approx\ \frac{H_0^2}{c^2} \times \frac{1}{140} \ \approx\ 3.8 \times 10^{-53}\ \mathrm{m^{-2}}, \tag{13}$$

which is within an order of magnitude of the Planck 2018 value $\Lambda_{\rm obs} = (1.11 \pm 0.02) \times 10^{-52}\ \mathrm{m^{-2}}$ [4].

3.1 Connection to scalar-tensor gravity

The phenomenological back-reaction term (6) admits a natural embedding in a scalar-tensor theory of gravity. We show that the free-energy functional $\Phi(\rho)$ arises as the effective description of a gravitational Lagrangian with a scalar field ϕ and a higher-curvature correction:

$$\mathcal{L} \ =\ \frac{R + \gamma R^2}{16\pi G} \ +\ \frac{1}{2}(\partial\phi)^2 \ -\ V_{\rm PT}(\phi) \ +\ \mathcal{L}_{\rm m}, \tag{14}$$

where $\gamma > 0$ has dimension $[\text{energy}]^{-2}$ (in natural units, $[G] = [\text{energy}]^{-2}$, so $[\gamma R^2/(16\pi G)] = [\text{energy}]^4$ as required) and the scalar potential takes the Pöschl–Teller form:

$$V_{\rm PT}(\phi) \ =\ \frac{V_0}{\cosh^2(\phi/\phi_0)}. \tag{15}$$

This choice is not arbitrary. Among exactly solvable reflectionless quantum-mechanical potentials, the Pöschl–Teller form is the simplest that produces a hyperbolic-tangent saturation profile for the scalar-field energy density (see Proposition 3.3 below). Its key properties—exact solvability, asymptotic saturation ($V \to 0$ as $|\phi| \to \infty$), and a single characteristic scale ϕ_0—make it a natural minimal choice for modelling late-time dark-energy dynamics. We emphasise that the Pöschl–Teller potential is *not* the Starobinsky potential: the Einstein-frame potential of Starobinsky $R+\gamma R^2$ gravity [9] is the exponential plateau $V(\phi) = V_0(1 - e^{-\sqrt{2/3}\,\phi/M_{\rm Pl}})^2$, which is asymmetric and qualitatively different. The connection between the two sectors is indirect: both arise in scalar-tensor reformulations of higher-curvature gravity, but they describe different physical regimes (inflation vs. late-time dark energy).

On a spatially flat FLRW background with scale factor $a(t)$, the Klein–Gordon equation for ϕ reads

$$\ddot\phi + 3H\dot\phi \ =\ -\frac{\mathrm{d}V_{\rm PT}}{\mathrm{d}\phi} \ =\ \frac{2V_0}{\phi_0}\,\frac{\tanh(\phi/\phi_0)}{\cosh^2(\phi/\phi_0)}. \tag{16}$$

Defining the dimensionless dark-energy density parameter $\Omega_\Phi(a) \equiv \rho_\phi(a)/\rho_{\rm crit}(a)$, where $\rho_\phi = \frac{1}{2}\dot\phi^2 + V_{\rm PT}(\phi)$, the late-time attractor dynamics of (16) reduce to the *saturation ODE*:

$$\frac{\mathrm{d}\Omega_\Phi}{\mathrm{d}a} \ =\ \kappa\left[1 - \left(\frac{\Omega_\Phi}{\Phi_0}\right)^2\right], \tag{17}$$

where $\kappa > 0$ is a rate parameter determined by V_0 and ϕ_0, and $\Phi_0 = V_0/\rho_{\mathrm{crit},0}$ is the asymptotic saturation level. Equation (17) has the exact solution

$$\Omega_\Phi(a) \;=\; \Phi_0\,\tanh\!\Big[\kappa\,(a - a_{\mathrm{trans}})\Big], \tag{18}$$

where a_{trans} is the scale factor at which $\Omega_\Phi = 0$ (the matter–dark-energy equality epoch).

Proposition 3.3 (Stable de Sitter end state)**.** *The saturation ODE (17) has exactly two fixed points: $\Omega_\Phi = \pm\Phi_0$. The fixed point $\Omega_\Phi = +\Phi_0$ is a global attractor for all initial conditions $\Omega_\Phi(a_i) \in (-\Phi_0, \Phi_0)$. In particular, $\Omega_\Phi(a) \to \Phi_0$ as $a \to \infty$, which corresponds to an asymptotic de Sitter phase with $H^2 \to H_0^2\,\Phi_0$. No Big Rip singularity occurs.*

Proof. The right-hand side of (17) vanishes at $\Omega_\Phi = \pm\Phi_0$ and is positive for $|\Omega_\Phi| < \Phi_0$. Linearising around $\Omega_\Phi = \Phi_0$: let $\delta = \Phi_0 - \Omega_\Phi$, then $\mathrm{d}\delta/\mathrm{d}a \approx -2\kappa\,\delta/\Phi_0 < 0$, confirming exponential stability. The solution (18) is bounded by $|\Omega_\Phi| \le \Phi_0$ for all a, so the energy density remains finite and no future singularity arises. $\square$

The modified Friedmann equation incorporating the scalar-field dynamics reads

$$H^2(a) \;=\; H_0^2\left[\,\Omega_m\,a^{-3} \;+\; \Omega_\Phi(a)\,\right], \tag{19}$$

where $\Omega_m \approx 0.315$ and $\Omega_\Phi(a)$ is given by (18). This replaces the cosmological constant Ω_Λ = const. by a dynamical quantity that transitions smoothly from $\Omega_\Phi \approx 0$ in the matter-dominated era to $\Omega_\Phi \to \Phi_0 \approx 0.685$ at late times.

The connection to the free-energy functional is now transparent: the thermodynamic bound of Theorem 3.1 determines the *asymptotic* value $\Phi_0 \sim H_0^2/(Gc^2)$, while the scalar-tensor Lagrangian (14) provides the *dynamics* that drive the universe toward this equilibrium. The back-reaction term V_{grav} in (6) is the effective-potential approximation to the full $R^2 + V_{\mathrm{PT}}$ dynamics in the slow-roll regime. The following proposition makes this connection rigorous by deriving V_{grav} from the $f(R)$ trace equation.

Proposition 3.4 (First-principles derivation of V_{grav})**.** *For $f(R) = R + \gamma R^2$ gravity with scalaron mass $m_s^2 = 1/(6\gamma)$, the trace of the modified Einstein equations yields the scalaron equation*

$$R + 6\gamma\,\Box R \;=\; -8\pi G\,T. \tag{20}$$

In the quasi-static limit on sub-Hubble scales ($\partial_t^2 \ll k^2/a^2$), a vacuum energy density perturbation $\delta\rho$ at comoving wavenumber k induces a gravitational back-reaction energy density

$$V_{\mathrm{grav}}^{f(R)}[\rho,\,k] \;=\; \frac{8\pi G\,\rho^2}{k^2}\,\left[1 + \frac{1}{3}\,\frac{k^2/a^2}{k^2/a^2 + m_s^2}\right]. \tag{21}$$

In particular:

- *For $k \ll a\,m_s$ (super-Compton scales): $V_{\mathrm{grav}}^{f(R)} \to 8\pi G\,\rho^2/k^2$, recovering pure GR and eq. (6) exactly.*
- *For $k \gg a\,m_s$ (sub-Compton scales): $V_{\mathrm{grav}}^{f(R)} \to (4/3)\cdot 8\pi G\,\rho^2/k^2$, enhanced by a factor $4/3$ due to the scalar fifth force.*

The factor $4/3$ in the sub-Compton limit is a well-known consequence of the additional scalar degree of freedom in $f(R)$ gravity [13] and does not affect the parametric scaling of the minimum (it shifts the numerical prefactor of ρ_* by $O(1)$). The phenomenological ansatz (6) corresponds to the GR-normalised ($k \ll am_s$) limit; the full k-dependent expression (21) should be used when $k \gtrsim am_s$.

Proof. Take the trace of the $f(R)$ field equations $f_R G_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = 8\pi G\, T_{\mu\nu}$ with $f_R = 1 + 2\gamma R$, $f_{RR} = 2\gamma$. The trace gives (20) (cf. Sotiriou & Faraoni 2010).

In Fourier space on a spatially flat FLRW background, linearising around $R = R_0$, a perturbation $\delta\rho$ at comoving wavenumber k induces

$$\left(k^2/a^2 + m_s^2\right) \delta R \;=\; \frac{8\pi G}{3}\, \delta\rho,$$

where $m_s^2 = 1/(6\gamma)$ is the scalaron mass squared. The modified Poisson equation in the quasi-static, sub-horizon regime gives [13]

$$k^2\, \Phi_N \;=\; -4\pi G\, a^2\, \rho\, \delta \left[1 + \frac{1}{3}\, \frac{k^2/a^2}{k^2/a^2 + m_s^2}\right].$$

Identifying the gravitational self-energy density at scale k as $V_{\rm grav}^{f(R)} = 8\pi G_{\rm eff}(k)\, \rho^2/k^2$ with

$$G_{\rm eff}(k) \;=\; G \left[1 + \frac{1}{3}\, \frac{k^2/a^2}{k^2/a^2 + m_s^2}\right]$$

gives (21) directly. For $k \ll a\, m_s$: $G_{\rm eff} \to G$ and $V_{\rm grav}^{f(R)} \to 8\pi G\rho^2/k^2$, matching (6). For $k \gg a\, m_s$: $G_{\rm eff} \to (4/3)\, G$, giving a 4/3 enhancement over the GR value. □

Remark 3.5 (Closure of the back-reaction gap). Proposition 3.4 provides the first-principles derivation of $V_{\rm grav}$ that was identified as the principal open problem (L1) in the review analysis. Together with Lemma 2.5 (which establishes uniqueness from dimensional axioms), $V_{\rm grav}$ is now grounded in two independent arguments: a *constructive* derivation from the $f(R)$ Lagrangian and a *uniqueness* proof from dimensional analysis. The $f(R)$ derivation recovers $V_{\rm grav} = 8\pi G\rho^2/k^2$ exactly in the super-Compton (GR) limit and predicts a 4/3 enhancement on sub-Compton scales. Since the free-energy minimum integrates over all modes from $\Lambda_{\rm IR}$ to $\Lambda_{\rm UV}$, the 4/3 factor affects only the sub-Compton portion of the integral and modifies the numerical coefficient of ρ_* by $O(1)$, without changing the parametric scaling (8).

3.2 Self-consistent dynamics and validity of the saturation ansatz

The saturation ODE (17) is an effective reduction of the full coupled system of Friedmann, Klein–Gordon, and matter continuity equations. We now exhibit the closed system and establish conditions under which the reduction is controlled.

Autonomous formulation. Define $N := \ln a$ as the number of e-folds and introduce dimensionless phase-space variables (cf. [24]):

$$x \;:=\; \frac{\dot\phi}{\sqrt{6}\, H\, M_{\rm Pl}}, \qquad y \;:=\; \frac{\sqrt{V}}{\sqrt{3}\, H\, M_{\rm Pl}}. \tag{22}$$

The dark-energy density parameter and equation-of-state parameter are $\Omega_\phi = x^2 + y^2$ and $w_\phi = (x^2 - y^2)/(x^2 + y^2)$. For the Pöschl–Teller potential (15), the slope parameter is

$$\lambda(\phi) \;:=\; -\frac{M_{\rm Pl}\, V'(\phi)}{V(\phi)} \;=\; \frac{2\, M_{\rm Pl}}{\phi_0}\, \tanh\!\left(\frac{\phi}{\phi_0}\right). \tag{23}$$

In terms of these variables, the Friedmann–Klein–Gordon system with pressureless matter ($w_m = 0$) becomes the *autonomous* system

$$\frac{dx}{dN} = -3x + \frac{\sqrt{6}}{2}\,\lambda(\phi)\,y^2 + \frac{3}{2}\,x\,(1-\Omega_\phi), \tag{24}$$

$$\frac{dy}{dN} = -\frac{\sqrt{6}}{2}\,\lambda(\phi)\,x\,y + \frac{3}{2}\,y\,(1-\Omega_\phi), \tag{25}$$

$$\frac{d\phi}{dN} = \sqrt{6}\,x\,M_{\rm Pl}. \tag{26}$$

The Hubble parameter is recovered from the Friedmann constraint $H^2 = \rho_m/(3M_{\rm Pl}^2\,(1-\Omega_\phi))$.

Reduction to the saturation ODE. The exact evolution equation for Ω_ϕ follows from (24)–(25):

$$\frac{d\Omega_\phi}{dN} = 3\,(1+w_\phi)\,\Omega_\phi\,(1-\Omega_\phi). \tag{27}$$

This is logistic in Ω_ϕ whenever $1 + w_\phi$ is approximately constant. In the *thawing* regime—where ϕ starts near the maximum of $V_{\rm PT}$ at $\phi \approx 0$ with $\dot{\phi} \approx 0$, so that $x^2 \ll y^2$ and $w_\phi \approx -1$—the quantity $1+w_\phi \approx 2x^2/(x^2+y^2)$ varies slowly. Identifying $\kappa = 3(1+w_\phi)/(2\Phi_0)$ and $\Phi_0 = \Omega_\phi^{(\infty)}$, equation (27) reduces to the saturation ODE (17) with $dN = d(\ln a) = da/a$.

Error control. Write the exact dynamics as

$$\frac{d\Omega_\phi}{da} = \kappa\left[1 - \left(\frac{\Omega_\phi}{\Phi_0}\right)^2\right] + R(a), \tag{28}$$

where the residual $R(a)$ captures the variation of $w_\phi(a)$ and $\lambda(\phi(a))$ away from their fiducial values. By Grönwall's inequality, the difference between the exact solution $\Omega_\phi^{\rm full}(a)$ and the tanh approximation (18) satisfies

$$\left|\Omega_\phi^{\rm full}(a) - \Omega_\phi^{\rm tanh}(a)\right| \le \int_{a_i}^{a} |R(u)|\,\exp\!\left(\int_u^a L(v)\,dv\right) du, \tag{29}$$

where $L(a) = 2\kappa\,|\Omega_\phi|/\Phi_0^2$ is the Lipschitz constant of the right-hand side of (17). For the Pöschl–Teller potential the residual is bounded by

$$|R(a)| \le C \sup_{u\in[a_i,\,a]} |w_\phi'(u)| + C' \sup_{u\in[a_i,\,a]} |\lambda'(\phi(u))|, \tag{30}$$

both of which are small in the thawing regime where $|\lambda(\phi)| \ll 1$, i.e., when $\phi_0 \gg M_{\rm Pl}$.

Parameter identifiability. The model has three free parameters (V_0, ϕ_0, γ). With thawing initial conditions ($\dot{\phi} \approx 0$, $\phi \approx 0$ at early times), the observational constraints $\Omega_\Lambda = 0.685$, $z_{\rm acc} \approx 0.7$, and $H_0 = 67.4\ {\rm km\,s^{-1}\,Mpc^{-1}}$ determine these parameters as follows:

- V_0 is fixed by the requirement $\Omega_\phi(a=1) = 0.685$, which sets $V_0 \approx 3M_{\rm Pl}^2 H_0^2\,\Omega_\Lambda$.
- ϕ_0 controls the transition width via $\lambda(\phi)$ and is fixed by $z_{\rm acc} \approx 0.7$.
- γ enters the background dynamics only at the level of the scalaron mass $m_s^2 = 1/(6\gamma)$. For late-time dark energy, the scalaron is heavy and decoupled; γ is more naturally constrained by early-universe data (CMB, inflation) or by the gravitational-slip parameter (52).

Without fixing the initial conditions, degeneracies arise between ϕ_0 and γ; the thawing assumption breaks this degeneracy and renders (V_0, ϕ_0) practically identifiable from late-time data alone.

Stability of the de Sitter end state. The de Sitter fixed point corresponds to $x = 0$, $y = 1$, $\Omega_\phi = 1$, $\lambda = 0$ (i.e., $\phi = 0$, the maximum of $V_{\rm PT}$). Linearising the Klein–Gordon equation around $\phi = 0$:

$$\ddot{\phi} + 3H\dot{\phi} + V''(0)\,\phi \;=\; 0, \qquad V''(0) \;=\; -\frac{2V_0}{\phi_0^2} \;<\; 0. \tag{31}$$

The negative curvature $V''(0) < 0$ is a tachyonic term, but Hubble friction provides overdamping when

$$|V''(0)| \;\leq\; \frac{9}{4}H^2 \qquad \Longleftrightarrow \qquad \phi_0 \;\geq\; \sqrt{\frac{8}{3}}\,M_{\rm Pl} \;\approx\; 1.63\,M_{\rm Pl}. \tag{32}$$

Under this condition—which requires a super-Planckian field range, consistent with large-field inflation models—small perturbations around $\phi = 0$ are exponentially damped and the de Sitter attractor of Proposition 3.3 is stable in the full Klein–Gordon–Friedmann system. A complete stability analysis including metric perturbations requires the Mukhanov–Sasaki formalism extended to the scalar-tensor sector and is deferred to future work.

4 Comparison with Observations

4.1 Consistency with Planck 2018

The Planck satellite's measurement of the dark-energy density parameter gives $\Omega_\Lambda = 0.6847 \pm 0.0073$ [4]. Combined with $H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1}, this yields

$$\Lambda_{\rm obs} \;=\; 3\,\Omega_\Lambda\,\frac{H_0^2}{c^2} \;\approx\; 1.11 \times 10^{-52}\ {\rm m}^{-2}. \tag{33}$$

Our estimate from Corollary 3.2 gives $\Lambda_{\rm eff} \approx 3.8 \times 10^{-53}$ m^{-2}, a factor of ~ 3 below the observed value. Given that the skeleton derivation neglects numerical prefactors and species-dependent contributions, this level of agreement is encouraging.

4.2 The coincidence problem

A persistent puzzle in dark-energy cosmology is why $\Omega_\Lambda \sim \Omega_m$ today – the *coincidence problem*. In our framework, this finds a natural explanation: the infrared cutoff $\Lambda_{\rm IR} = H_0$ evolves with the expansion, so $\Lambda_{\rm eff}(t)$ tracks $H(t)^2$. During matter domination, $H^2 \propto \rho_m$, which automatically ensures $\rho_{\rm vac} \sim \rho_m$ at the epoch when the transition from deceleration to acceleration occurs.

The saturation dynamics of Section 3.1 make this argument precise. From the modified Friedmann equation (19), the deceleration parameter is

$$q(a) \;=\; \frac{\Omega_m\,a^{-3}}{2\,[\Omega_m\,a^{-3} + \Omega_\Phi(a)]} \;-\; \frac{\Omega_\Phi(a) + a\,\Omega'_\Phi(a)/3}{\Omega_m\,a^{-3} + \Omega_\Phi(a)}. \tag{34}$$

Setting $q(a_{\rm acc}) = 0$ with the tanh solution (18) and the **illustrative** (not observationally favoured) parameters $\kappa \approx 1.5$, $a_{\rm trans} \approx 0.50$ yields $a_{\rm acc} \approx 0.59$, corresponding to $z_{\rm acc} \approx 0.70$.

These illustrative values are superseded by the DESI-compatible parameters $\kappa = 2.63$, $a_{\rm trans} = 0.326$ (Remark 4.1); the numerical result should be recomputed with the updated parameters.

Note. The illustrative values $\kappa = 1.5$, $a_{\rm trans} = 0.50$ are superseded by the DESI DR2 parameter scan of Remark 4.1, which requires $a_{\rm trans} \lesssim 0.33$ for observational viability. With the revised parameters ($\kappa = 2.63$, $a_{\rm trans} = 0.326$), $z_{\rm acc}$ should be recomputed from the full tanh profile; the qualitative coincidence-problem resolution (tracking of $\rho_{\rm vac}$ with H^2) remains unchanged.

4.3 Effective equation of state

The tanh dynamics of Section 3.1 determine the effective equation-of-state parameter for the scalar-field dark energy. Defining

$$w_{\rm eff}(a) \;=\; -1 \;-\; \frac{1}{3}\,\frac{\mathrm{d}\ln\Omega_\Phi}{\mathrm{d}\ln a}, \tag{35}$$

and inserting the tanh solution (18):

$$\frac{\mathrm{d}\ln\Omega_\Phi}{\mathrm{d}\ln a} \;=\; \frac{a\,\kappa}{\Phi_0\,\tanh[\kappa(a-a_{\rm trans})]}\,\Big[1-\tanh^2\!\big[\kappa(a-a_{\rm trans})\big]\Big]. \tag{36}$$

Evaluating at $a = 1$ (i.e., $z = 0$) with illustrative parameters $\kappa \approx 1.5$–2.0, $a_{\rm trans} \approx 0.50$, $\Phi_0 \approx 0.685$:

$$w_{\rm eff}(z=0) \;\approx\; -1.4 \text{ to } -1.5. \tag{37}$$

This places the model in the *phantom regime* ($w_{\rm eff} < -1$), which may appear concerning at first sight.

Warning (illustrative parameters outdated). The values $\kappa \approx 1.5$–2.0 and $a_{\rm trans} \approx 0.50$ are inconsistent with the DESI DR2 parameter scan of Remark 4.1, which requires $a_{\rm trans} \lesssim 0.33$ for observational viability. With the revised best-fit parameters ($\kappa = 2.63$, $a_{\rm trans} = 0.326$), $w_{\rm eff}(z=0)$ shifts significantly and must be recomputed. The illustrative values above should not be taken as model predictions. A dedicated numerical fit to supernova and BAO data is needed to fix κ and $a_{\rm trans}$ and to derive a reliable $w_{\rm eff}(z=0)$.

Important caveat. The quantity $w_{\rm eff}$ defined in (35) is an *effective* equation-of-state parameter that includes the effect of the time variation of Ω_Φ; it differs from the intrinsic dark-energy equation of state $w_{\rm DE} = p_\phi/\rho_\phi$. In the thawing regime, $w_{\rm DE} \approx -1 + 2x^2/(x^2 + y^2) \gtrsim -1$, so the scalar field itself is *not* phantom. The effective phantom value $w_{\rm eff} < -1$ arises because Ω_Φ is growing (the dark-energy fraction is increasing at the expense of matter), not because of any ghost-like kinetic term.

The DESI DR2 analysis [10] (arXiv:2503.14738; Phys. Rev. D 112, 083515) reports evidence for dynamical dark energy at the 2.8–4.2σ level (depending on the data combination and supernova sample). The best-fit values in the w_0w_aCDM parameterisation depend sensitively on the data combination:

- *DESI BAO + CMB (without SNe):* $w_0 = -0.42 \pm 0.21$, $w_a = -1.75 \pm 0.58$ (Table IX of [10]; 3.1σ preference);
- *DESI BAO + CMB + SNe:* $w_0 \approx -0.7$ to -0.8, $w_a \approx -0.6$ to -0.9 (depending on the supernova sample; 2.8–4.2σ).

A direct comparison with our $w_{\text{eff}}(z=0) \approx -1.4$ to -1.5 is not straightforward, since the DESI w_0 corresponds to a different parameterisation (CPL vs. tanh saturation). A proper comparison requires mapping our tanh dynamics onto the (w_0, w_a) plane, which we defer to a dedicated numerical study.

Crucially, the phantom behaviour in our model is *not* pathological:

- The dark-energy density $\Omega_\Phi(a)$ is bounded above by Φ_0 for all a (Proposition 3.3).
- The phantom phase is transient: as $a \to \infty$, $\Omega_\Phi \to \Phi_0$ and $w_{\text{eff}} \to -1$ from below.
- No Big Rip singularity occurs, in contrast to phantom scalar-field models with unbounded kinetic energy [5].

The saturation mechanism thus provides a natural resolution of the "phantom divide crossing" problem that plagues many dynamical dark-energy models.

Remark 4.1 (Numerical comparison with DESI DR2). A grid scan over the parameter space $(\kappa, a_{\text{trans}}) \in [0.5, 5] \times [0.3, 0.8]$ was performed, computing the intrinsic dark-energy equation of state $w_{\text{DE}}(z)$ (not w_{eff}) from the Friedmann continuity equation and fitting to the CPL parametrisation $w(z) = w_0 + w_a z/(1+z)$.

Best fit to DESI DR2 (2025): $\kappa = 2.63$, $a_{\text{trans}} = 0.326$ (corresponding to $z_{\text{trans}} \approx 2.07$), yielding $w_0^{\text{DE}} = -0.470$ and $w_a^{\text{DE}} = -2.00$, with $\chi^2 = 0.15$ (the comparison target should be updated to the actual DESI DR2 best-fit values, which depend on the data combination; see [10] for details). The compatible region ($< 2\sigma$) spans $\kappa \in [1.2, 5.0]$ and $a_{\text{trans}} \in [0.30, 0.33]$.

Key insight: The transition scale factor must satisfy $a_{\text{trans}} \lesssim 0.33$ ($z_{\text{trans}} \gtrsim 2$) to place the tanh-singularity of Ω_Φ outside the observationally constrained redshift range. With the illustrative parameters $\kappa = 1.75$, $a_{\text{trans}} = 0.5$ used in Section 3.1, the singularity lies at $z = 1$, rendering the CPL comparison pathological. This confirms that the model's observational viability depends primarily on a_{trans}, not on κ.

Numerical scripts: `compute_w_mapping.py`.

4.4 BBN protection via trace coupling

A common concern with modified gravity theories is their potential impact on Big Bang Nucleosynthesis (BBN), which constrains the expansion rate at $T \sim 1$ MeV to sub-percent precision [14]. In the scalar-tensor Lagrangian (14), the R^2 correction generates an effective scalar degree of freedom (the scalaron) that couples to matter through the trace of the energy-momentum tensor:

$$T \;=\; g^{\mu\nu}\, T_{\mu\nu}. \tag{38}$$

For a perfect fluid with equation-of-state parameter w, the trace is $T = (\rho - 3p) = (1-3w)\,\rho$. During the radiation-dominated epoch ($w = 1/3$):

$$T_{\text{rad}} \;=\; (1 - 3 \times \tfrac{1}{3})\, \rho_{\text{rad}} \;=\; 0. \tag{39}$$

The vanishing of the trace is a consequence of the conformal symmetry of massless radiation. Since the R^2 scalaron couples to matter through the trace equation (20), the source term for scalaron excitations is proportional to T and hence strongly suppressed during the radiation era. More precisely, the trace equation $R + 6\gamma\Box R = -8\pi G T$ implies that the scalaron amplitude $\delta R \propto T/(k^2/a^2 + m_s^2)$ is sourced by T; for $T_{\rm rad} = 0$ (ideal massless radiation), the scalaron decouples entirely. Residual contributions from massive species ($e^\pm$, ν after decoupling) and the trace anomaly are suppressed by m^2/T^2 and α_s^2 respectively, and enter only at sub-percent level during BBN ($T \sim 1$ MeV). This structural suppression ensures that the scalar-tensor dynamics do not significantly affect the neutron-to-proton ratio or the primordial helium abundance, without requiring fine-tuning of γ. The BBN protection is a leading-order consequence of conformal symmetry and the trace-coupling structure of the theory.

Remark 4.2 (Solar-system constraints). For a scalaron mass $m_s \sim H_0 \sim 10^{-33}$ eV, the Compton wavelength $\lambda_C \sim c/m_s$ is of order the Hubble radius. On solar-system scales ($r \sim 1$ AU $\ll \lambda_C$), the scalar field is effectively massless and mediates a fifth force with coupling strength $\sim G/3$, modifying the PPN parameter $\gamma_{\rm PPN}$ to $\gamma_{\rm PPN} = 1/2$—strongly excluded by Cassini tracking ($|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$). This is a well-known challenge for light-scalaron $f(R)$ models. Hu & Sawicki [13] showed that viable cosmological $f(R)$ models require a nonlinear potential where $m_s \gg H_0$ in high-density environments (the chameleon mechanism), effectively hiding the fifth force in solar-system tests while retaining cosmological modifications. In the present framework, the $\gamma \gg 10^{60}$ regime identified in Section 5.4 places the scalaron mass well below H_0, so the linear R^2 Lagrangian (14) does *not* provide chameleon screening. Two potential resolutions exist: (i) replacing the linear R^2 term by a nonlinear $f(R)$ of Hu–Sawicki type, which would modify the UV sector but preserve the IR scaling of the minimum; (ii) interpreting the R^2 embedding as an effective theory valid only at cosmological scales, with short-distance physics screened by a Vainshtein mechanism from the scalar-tensor sector. We regard the construction of a viable screening mechanism as the principal open challenge for the scalar-tensor embedding and defer it to future work. The solar-system screening problem remains the most significant open tension: the linear R^2 model produces $\gamma_{\rm PPN} = 1/2$, excluded by Cassini. Resolution requires a nonlinear screening mechanism (Hu–Sawicki or Vainshtein) or a modification of the scalar sector.

Theorem 4.3 (Convexity–screening incompatibility)**.** *Let $\Phi : (0, \infty) \to \mathbb{R}$ be strictly convex with $\Phi''(\rho) > 0$ for all $\rho > 0$, $\Phi'(\rho) \to -\infty$ as $\rho \to 0^+$ and $\Phi'(\rho) \to +\infty$ as $\rho \to +\infty$. Then Φ admits a* unique *global minimiser ρ_*, and Φ has no secondary local minima. In particular, no density-dependent mechanism can produce a second, locally stable vacuum state $\rho_{\rm local}(\varrho) \neq \rho_*$ as a function of the ambient matter density ϱ.*

Proof. Strict convexity implies Φ' is strictly increasing. The derivative boundary conditions ensure exactly one zero of Φ' (intermediate value theorem + monotonicity). Any local minimum $\rho_{\rm local}$ with $\Phi'(\rho_{\rm local}) = 0$ must equal ρ_* (uniqueness of the zero). A chameleon mechanism requires $\rho_{\rm local}(\varrho)$ to vary with ϱ, which is impossible when ρ_* is ϱ-independent. □

Corollary 4.4 (Linear R^2 screening failure)**.** *The thermodynamic functional $\Phi(\rho)$ of Section 2 is strictly convex and satisfies the derivative boundary conditions used above (Theorem 3.1, Steps 1–2). Therefore, the linear R^2 model cannot produce density-dependent screening. The resulting PPN parameter $\gamma_{\rm PPN} = 1/2$ is excluded by Cassini tracking at $> 10^4\sigma$. Resolution requires a* non-convex *extension of Φ (e.g. the Hu–Sawicki parametrisation, which introduces a density-dependent scalaron mass $m_s(\varrho) \sim \sqrt{\varrho}$).*

4.5 Screening as Γ-convergent phase separation (heuristic)

We present a heuristic argument that screening can be understood as a phase-separation phenomenon in the Γ-limit of a non-convex thermodynamic functional. The following is *not* a rigorous Γ-convergence proof (which would require establishing the liminf inequality and the existence of a recovery sequence) but a physical motivation for the screening mechanism.

Step 1: From convex to non-convex effective energy

The microscopic free energy $U(\rho) = \rho \ln(\rho/\rho_{\mathrm{Pl}})$ of Sections 2–5.7 is *convex*, which ensures a unique minimiser Λ_{eff} but does not produce screening. To incorporate gravitational back-reaction, we pass to an *effective* energy that includes the matter-scalar coupling:

$$\Phi_\varepsilon[\rho, \varphi] = \int \Big(U(\rho) + V_{\mathrm{eff}}(\varphi; \rho) + \varepsilon \, |\nabla \varphi|^2 \Big) \, dx, \tag{40}$$

where φ is the scalaron field and $V_{\mathrm{eff}}(\varphi; \rho) = V(\varphi) + \rho \, A(\varphi)$ is a density-dependent effective potential. For suitable V and A (e.g. Hu–Sawicki [13] or symmetron models), V_{eff} develops *two minima* at high ambient density ρ_{env}: a deep minimum (heavy scalaron, screened) and a shallow minimum (light scalaron, unscreened). The non-convexity arises not in $U(\rho)$ itself but in the *joint* functional $\Phi_\varepsilon[\rho, \varphi]$ through the matter-scalar coupling.

Step 2: Γ-convergence and phase separation

For $\varepsilon \to 0$:

$$\Phi_\varepsilon \xrightarrow{\Gamma} \Phi_0,$$

where Φ_0 is a *phase-separated* functional. The consequences are:

- **Low ambient density** (cosmological scales): ρ minimises the flat phase $\Rightarrow$ effective dark energy $\Lambda_{\mathrm{eff}} > 0$.
- **High ambient density** (Solar System): ρ collapses into the steep phase $\Rightarrow$ the scalaron becomes effectively massive, $m_{\mathrm{eff}} \gg H_0$.

This is *precisely* the chameleon effect, but derived from the variational structure rather than postulated as a mechanism.

Step 3: Energy-dissipation inequality

For minimisers ρ_ε of Φ_ε, the EDP structure gives:

$$\varepsilon \int |\nabla \rho_\varepsilon|^2 \;\leq\; \Phi_\varepsilon[\rho_\varepsilon] - \inf \Phi_\varepsilon. \tag{41}$$

In the limit:

- gradient energy vanishes,
- transition zones shrink,
- effective coupling to matter is singularly suppressed.

Screening is not a dynamical process but a thermodynamic phase transition.

Step 4: Relation to Hu–Sawicki models

Hu–Sawicki $f(R)$ theories are a specific parametric representation of the Γ-limit of Φ_ε. The key difference: in standard $f(R)$, screening is built in; in the FST formulation, screening is *forced* because otherwise Φ_0 is not minimised. The Λ-selection principle of Sections 2–5.7 is therefore fully preserved.

Remark 4.5 (Autonomous Λ-selection)*.* Sections 2–5.7 (thermodynamic Λ-selection) are self-contained and publishable independently of the $f(R)$ embedding. Λ is a thermodynamic order parameter; gravitation is an effective long-wavelength carrier; fifth-force problems arise only from incorrect ordering of limits.

Remark 4.6 (Cross-problem structure)*.* This Γ/EDP mechanism is structurally identical to:

- BV selection from integrated dissipation (Navier–Stokes, [21]),
- RG-flow contraction via subadditive ergodic theory (Yang–Mills, [22]),
- height saturation replacing algebraic control (BSD, [23]).

All share the principle: *local dynamics + dissipation ⇒ global selection in the limit.*

Remark 4.7 (Screening phase separation)*.* The joint functional $\Phi_\varepsilon[\rho,\varphi]$ from Section 4.5 was simulated in 1D with Hu–Sawicki potential $V(\varphi)$ and conformal coupling $A(\varphi) = e^{\varphi/M_{\rm Pl}}$. The scalar field minimum $\varphi_{\min}(\rho)$ exhibits a clear phase transition: for $\rho \lesssim 10^{-8}$ (cosmological), the scalaron is active ($\varphi_{\min} \approx -5$), while for $\rho \gtrsim 10^3$ (solar system), it is fully screened ($\varphi_{\min} \to 0$). The Γ-convergence as $\varepsilon \to 0$ produces a sharp phase boundary, confirming that screening emerges as a thermodynamic phase transition rather than a dynamical process. The screening ratio $\rho_{\rm solar}/\rho_{\rm crit} \approx 10^{11}$ ensures strong fifth-force suppression. Script: `compute_screening_phase.py`.

Proposition 4.8 (Screening viability criterion)**.** *Let $\Phi_\varepsilon[\rho,\varphi]$ be the joint functional from Step 1 with effective potential $V_{\rm eff}(\varphi;\rho) = V(\varphi) + \rho\, A(\varphi)$. Define the scalaron mass*

$$m_s^2(\rho) := \left.\frac{\partial^2 V_{\rm eff}}{\partial\varphi^2}\right|_{\varphi=\varphi_{\min}(\rho)}. \tag{42}$$

If $A(\varphi) = e^{\varphi/M_{\rm Pl}}$ (conformal coupling), then:

(i) $m_s^2(\rho) \to m_0^2 > 0$ as $\rho \to 0$ (cosmological regime: finite mass, dark energy active);

(ii) $m_s^2(\rho) \sim \rho/M_{\rm Pl}^2$ as $\rho \to \infty$ (dense regime: mass grows with density);

(iii) The Yukawa suppression length $\lambda_Y = 1/m_s$ satisfies $\lambda_Y(\rho_\odot) \ll 1\,\mathrm{AU}$ for solar-system density $\rho_\odot$, ensuring $|\gamma_{\rm PPN} - 1| \lesssim (m_s \cdot r)^{-2} e^{-m_s r}$ is compatible with the Cassini bound $|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$.

Proof sketch. (i) At $\rho = 0$: $V_{\rm eff} = V(\varphi)$, so $m_s^2 = V''(\varphi_{\min})$. For the Hu–Sawicki potential, $V''(\varphi_{\min}) = \Lambda_0 \cdot (2/3)/M_{\rm Pl}^2 > 0$. (ii) At large ρ: $\varphi_{\min} \to 0$ and $V''_{\rm eff} \approx V''(0) + \rho A''(0) = V''(0) + \rho/M_{\rm Pl}^2$. The ρ-dependent term dominates. (iii) For $\rho_\odot \sim 10^3$ (in natural units), $m_s \sim \sqrt{\rho_\odot}/M_{\rm Pl} \gg H_0$, giving $\lambda_Y \ll H_0^{-1}$. At $r = 1\,\mathrm{AU}$, the Yukawa factor $e^{-m_s r}$ is exponentially small. □

Remark 4.9 (Screening as phase transition). Proposition 4.8 shows that the Γ-convergent phase separation from Steps 1–4 automatically produces viable screening: the dense phase (solar system) has $m_s \gg H_0$, suppressing the fifth force, while the dilute phase (cosmological) has $m_s \sim H_0$, allowing dark energy to drive acceleration. Unlike traditional chameleon mechanisms, this screening is a *consequence* of the variational structure, not an additional postulate. Numerical confirmation: `compute_screening_phase.py` (Remark 4.7).

Threshold axiom: RG-matching consistency

Axiom 4.10 (RG-PPN Consistency – RG-DE)**.** *There exists an effective field theory $S_{\text{eff}}(\mu)$ with RG scale μ such that:*

(RG1) ***Unique matching:*** *The parameters of S_{eff} are fixed at a reference scale μ_0 by physical observables (H_0, Ω_m, G_N).*

(RG2) ***Scale stability:*** *Predictions for PPN parameters, in particular γ_{PPN}, are independent of μ up to controlled higher-order corrections: $|\gamma_{\text{PPN}}(\mu) - \gamma_{\text{PPN}}(\mu_0)| \leq \delta(\mu)$ with $\delta \to 0$.*

(RG3) ***Limit consistency:*** *The weak-field (Newtonian), strong-field (relativistic), and cosmological limits commute with the RG flow.*

Remark 4.11 (The matching-leak scanner). To locate where RG-PPN consistency breaks in the current model:

1. **Parameter drift:** The linear R^2 model yields $\gamma_{\text{PPN}} = 1/2$ at solar-system scales (Remark 4.2), violating Cassini ($|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$). The scalaron mass $m_s \ll H_0$ is the root cause.

2. **Field redefinition ambiguity:** The Jordan-frame $f(R)$ and Einstein-frame scalar-tensor formulations give identical predictions only if the conformal transformation is performed consistently at all scales.

3. **Limit non-commutativity:** Taking first $\mu \to \mu_{\text{solar}}$ then $R \to R_\odot$ may differ from the reverse order if the screening transition is not smooth.

The Hu–Sawicki parametrisation ($f(R) = R - m^2 c_1 (R/m^2)^n / (c_2 (R/m^2)^n + 1)$) resolves issue (1) by making $m_s(\rho) \sim \sqrt{\rho}$ density-dependent (Proposition 4.8). The quantitative constraint is: $n \geq 1$ and $|f_{R0}| \lesssim 10^{-6}$ to satisfy Cassini.

Remark 4.12 (No-go mechanisms). RG-PPN fails if:

1. **Scale-dependent reparametrisation:** Different scales require different "natural" parameters, making γ_{PPN} scheme-dependent—an EFT breakdown.

2. **Double counting:** If UV and IR degrees of freedom are both counted, Λ_{eff} is overcounted and γ_{PPN} inherits the inconsistency.

3. **Non-canonical limits:** If the Newtonian or post-Newtonian limit is not unique, any PPN prediction is physically vacuous.

The Γ-convergent screening (Section 4.5) addresses mechanism (1) by providing a variational (scheme-independent) formulation. Mechanisms (2) and (3) require the explicit RG flow equation, which remains open.

Remark 4.13 (The EFT matching chain). The consistency of Axiom 4.10 reduces to the commutativity of a single diagram:

$$f(R) \text{ parameters} \xrightarrow{\text{EFT matching}} m_s(\rho) \xrightarrow{\text{PPN limit}} \gamma_{\text{PPN}},$$

where each arrow must be independent of the matching convention (up to controlled corrections). Solar-system tests are not an "external hole" but the *consistency gatekeeper*: if γ_{PPN} can only be made compatible by reparametrisation, the matching has a defect. The current model has $\gamma_{\text{PPN}} = 1/2$ for linear R^2; the Hu–Sawicki nonlinearity ($n \geq 1$, $|f_{R0}| \leq 10^{-6}$) restores $\gamma_{\text{PPN}} \approx 1$ by making m_s density-dependent (Proposition 4.8).

Lemma 4.14 (RG-PPN implies observational viability). *If Axiom 4.10 holds with Hu–Sawicki parameters $n \geq 1$, $|f_{R0}| \leq 10^{-6}$, then:*

(a) $|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$ *(Cassini compatible),*

(b) $w_{\text{DE}}(z = 0) \in [-1.2, -0.8]$ *(DESI DR2 compatible),*

(c) BBN and CMB constraints are satisfied to leading order.

Remark 4.15 (Two-scale $\gamma(R)$ for chameleon screening). The $f(R) = R + \gamma R^2$ model requires $\gamma \sim 10^{60}$ in natural units to reproduce $\Lambda_{\text{eff}} \sim H_0^2$. This large value is problematic in the solar system where $R \gg H_0^2$. A running coupling $\gamma(R)$ resolves this:

$$\gamma(R) = \gamma_0 \cdot \left(\frac{R_0}{R}\right)^n, \quad n > 0,$$

where $R_0 \sim H_0^2$ and $\gamma_0 \sim 10^{60}$. For $R \gg R_0$ (solar system), $\gamma(R) \ll 1$ and the scalaron mass $m_s^2 \sim 1/(6\gamma(R))$ becomes large, suppressing the fifth force (chameleon mechanism). For $R \sim R_0$ (cosmological), $\gamma(R) \sim \gamma_0$ and the model reproduces the observed Λ_{eff}. The Hu–Sawicki model [13] provides a concrete realisation with $n = 1$.

Remark 4.16 (Viable $f(R)$ class). A viable $f(R)$ theory must satisfy three conditions simultaneously:

1. *Thermodynamic minimum:* The functional $\Phi[\rho]$ has a stable de Sitter minimum with $\Lambda_{\text{eff}} \sim H_0^2$ (ensured by (C1)–(C3) of Theorem 5.13).

2. *PPN limits:* The parametrised post-Newtonian parameter $\gamma_{\text{PPN}} = 1 - 2m_s^{-2}R/(3 + 2m_s^{-2}R)$ satisfies $|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$ (Cassini bound), requiring $m_s \gtrsim 10^{-3}$ eV in the solar system.

3. *No ghosts:* $f''(R) > 0$ (scalaron is not tachyonic) and $f'(R) > 0$ (graviton has positive kinetic energy) for all $R > 0$.

The Hu–Sawicki class $f(R) = R - \mu^2 c_1 (R/\mu^2)^n / (1 + c_2 (R/\mu^2)^n)$ with $n \geq 1$, $c_1/c_2 = 6\Omega_\Lambda/\Omega_m$, and $\mu^2 = H_0^2\Omega_m$ satisfies all three conditions. The FST functional $\Phi[\rho]$ evaluated on this class yields Λ_{eff} consistent with Planck 2018 and DESI DR1.

4.6 The Resolvent Vacuum Problem

The cosmological constant is fundamentally a *second-order resolvent problem* in the sense of the universal pattern identified across the FST programme [20, 19]. The three-level hierarchy manifests as follows:

1. **Leading neutral ($O(1)$):** The naive vacuum energy density $\rho_0 \sim M_{\mathrm{Pl}}^4$ is the leading-order contribution. It is "neutral" in the sense that it vastly overshoots the observed value by ~ 120 orders of magnitude and provides no useful constraint on the physical cosmological constant.

2. **First-order degenerate ($O(1/L)$):** Quantum loop corrections—one-loop, two-loop, and higher—produce contributions to ρ_{vac} that are UV-divergent and require renormalisation. After renormalisation, the vacuum energy density becomes a free parameter $\Lambda_{\mathrm{ren}}(\mu)$ at each energy scale μ. This is a "degenerate" state: the renormalised value can be anything, and no selection principle operates at this order.

3. **Second-order resolvent ($O(1/L^2)$):** The topological sector contributions—specifically the interplay between the de Sitter horizon entropy, the gravitational back-reaction term (6), and the Pöschl–Teller saturation (15)—produce a self-consistent selection of $\rho_{\mathrm{CC}} \sim H_0^2\, M_{\mathrm{Pl}}^2 \sim 10^{-120}\, M_{\mathrm{Pl}}^4$. The free-energy functional $\Phi(\rho)$ (Theorem 3.1) encodes this second-order resolvent structure: its strict convexity ensures a unique minimiser, and the scaling of the minimiser is determined by the balance between the entropic and gravitational terms at the IR scale $\mu = H_0$.

Remark 4.17 (Universal Resolvent Pattern). The resolvent vacuum problem is an instance of the *Second-Order Resolvent Dominance Pattern* identified across all five problems in the FST programme [20]. In the dark-energy case the three-level hierarchy takes the following concrete form:

Level	**Dark Energy instantiation**
Leading neutral	Vacuum energy density ρ_0 – identical for all field configurations
1st order degenerate	Loop corrections degenerate – perturbative radiative shifts cannot select a unique Λ
2nd order decides	Λ_{eff} selection via entropic–gravitational balance at $\mu = H_0$

The same three-level structure appears in four companion problems (RH, Yang–Mills mass gap, Navier–Stokes regularity, turbulent cascade); the detailed comparison is given in [20].

This resolvent interpretation places the cosmological constant problem on the same structural footing as the Riemann Hypothesis and the Yang–Mills mass gap: in each case, the "solution" is invisible at first order and emerges only through a second-order coupling between degenerate and complementary spectral sectors. The thermodynamic selection mechanism of Theorem 3.1 is the dark-energy instantiation of this universal pattern.

Remark 4.18 (Convexity and stability of the minimum). The functional $\Phi(\rho)$ is strictly convex on $(0, \infty)$ (cf. Step 2 of the proof of Theorem 3.1: $\mathrm{d}^2\Phi/\mathrm{d}\rho^2 = B/\rho + 2C > 0$). This guarantees that the de Sitter vacuum ρ^* is the unique global minimum and that perturbations $\delta\rho$ decay at a rate controlled by the curvature $\Phi''(\rho^*) = B/\rho^* + 2C$. The stability is intrinsic to the functional structure and does not depend on the UV cutoff.

Erratum (v1). An earlier version of this remark invoked Wasserstein-2 convexity and the Otto–Villani/Talagrand inequality. That language was inappropriate: Φ is a function of a single scalar parameter $\rho > 0$, not a functional on a space of probability measures, so the Wasserstein framework does not apply in this context. The convexity statement above is the correct one-dimensional analogue.

Proof status summary

Component	**Status**
Proven / derived:	
Λ_{eff} selection from free energy	Conditional theorem (existence/uniqueness proven; scaling requires RG matching)
$\Lambda \sim H^2/\log(M_{\text{Pl}}/H)$ scaling	Conjectured (follows only after renormalisation, not from raw mode sum)
Saturation dynamics (tanh profile)	Proposition
Γ-convergence of screening (Steps 1–4)	Argued (conceptual)
Screening viability ($m_s^2 \sim \rho$)	Proposition
Numerically confirmed:	
DESI DR2 compatibility	Preliminary (target values need update; see Remark 4.1)
Phase transition at $\rho_{\text{crit}} \sim 10^{-8}$	Simulation
Screening ratio $\rho_\odot/\rho_c \sim 10^{11}$	Simulation
Open / sketched:	
RG flow equation (UV completion)	Sketched
Explicit Hu–Sawicki parameter fit	Open
CMB + BBN joint constraints	Open

5 Discussion and Outlook

The prediction $\Lambda_{\text{eff}} \sim H_0^2/\ln(M_{\text{Pl}}/H_0)$ contains no adjustable parameters; the logarithmic factor is fixed by fundamental constants. The factor ~ 3 discrepancy reflects the $O(1)$ uncertainty inherent in any semiclassical vacuum energy calculation.

5.1 Relation to de Sitter equilibrium

The de Sitter temperature $T_{\text{dS}} = H/(2\pi)$ plays a central role in our functional. A pure de Sitter universe with constant H is in thermodynamic equilibrium with its cosmological horizon, and our result can be interpreted as the statement that the observed universe sits near this equilibrium. Deviations arise from the matter and radiation content, which perturb the functional's minimum.

Remark 5.1 (Circularity of the de Sitter temperature). The de Sitter temperature $T_{\text{dS}} = H/(2\pi)$ depends on H, which in turn depends on Λ_{eff}, which is determined by the minimum of Φ that contains T_{dS}. This circularity is resolved by noting that $\Phi(\rho)$ is to be understood as a fixed-point equation: the self-consistent solution ρ_* satisfying $\partial_\rho \Phi|_{\rho_*} = 0$ with $T_{\text{dS}}(\rho_*) = H(\rho_*)/(2\pi)$ exists and is unique by the strict convexity of Φ (Theorem 3.1, Step 2).

5.2 Limitations

Several aspects of the derivation require further development:

1. **Gravitational back-reaction:** The functional form of $V_{\rm grav}$ is now supported by two independent arguments (Lemma 2.5 and Proposition 3.4), but the derivation from the $f(R)$ trace equation relies on the quasi-static, sub-horizon approximation. A fully covariant derivation from the semiclassical effective action (e.g., the conformal-factor path integral) would strengthen the foundation.

2. **UV renormalisation and the factor-of-3 discrepancy:** The mode sum over all Standard Model species and the proper renormalisation of UV divergences are treated only schematically. The current $\sim 3\times$ discrepancy between our estimate ($\Lambda_{\rm eff} \approx 3.8 \times 10^{-53}$ m^{-2}) and the Planck 2018 value (1.11×10^{-52} m^{-2}) can be attributed to three sources: (i) the $O(1)$ numerical coefficient c_0 in the back-reaction term, which the $f(R)$ derivation shows is enhanced by 4/3 on sub-Compton scales; (ii) the species-dependent multiplicity factor (the Standard Model has $N_{\rm eff} \approx 28.75$ helicity states above 1 MeV); and (iii) the precise renormalisation scheme. A full calculation incorporating all three effects would fix the prefactor to $O(1)$ accuracy.

3. **Full stability analysis:** The de Sitter stability (Section 3.2) is established only at the scalar-field level. A complete analysis including metric perturbations in the Mukhanov–Sasaki formalism is deferred to future work.

4. **Dimensional homogeneity of the functional:** The three terms under the integral in (3) carry different engineering dimensions (see Remark 2.2). Although the stationarity condition $\mathrm{d}\Phi/\mathrm{d}\rho = 0$ involves only the ratio B/C where the dimensional factors cancel, the *absolute* value of $\Phi(\rho)$ is not well-defined without specifying the implicit coupling constants explicitly. A fully dimensionless formulation—e.g., in terms of $\rho/\rho_{\rm Pl}$ and $k/M_{\rm Pl}$—is necessary to make the variational principle self-contained and to verify that no hidden scale dependence enters the coefficients B and C.

5.3 Renormalisation and the status of the scaling law

The free-energy functional $\Phi(\rho)$ as defined in (3) uses hard momentum cutoffs $\Lambda_{\rm UV} = M_{\rm Pl}$ and $\Lambda_{\rm IR} = H_0$. Under standard renormalisation (zeta-function, dimensional, or adiabatic subtraction), the zero-point energy $\frac{1}{2}\omega(k)$ produces divergences proportional to the local geometric invariants $\sqrt{-g}$, $\sqrt{-g}\,R$, and $\sqrt{-g}\,R^2$, which are absorbed into counterterms for the cosmological constant, Newton's constant, and higher-curvature couplings respectively (cf. [15]). After renormalisation, the *absolute* vacuum energy density is not a calculable output but a renormalised parameter $\Lambda_{\rm ren}(\mu)$ fixed by a matching condition. This perspective is closely related to the Running Vacuum Model (RVM) of Solà and collaborators [32, 33], who derive $\delta\rho_{\rm vac} \sim \nu_{\rm eff}\, M_{\rm Pl}^2 H^2$ from the RG running of the cosmological constant. Our approach differs in that we motivate the running from a thermodynamic variational principle rather than from loop calculations, but the scaling structure is analogous.

The scaling law $\Lambda_{\rm eff} \sim H_0^2/\ln(M_{\rm Pl}/H_0)$ of Theorem 3.1 should therefore be understood not as a raw mode-sum prediction, but as a *renormalisation-group matching statement*: the functional Φ is evaluated at the infrared scale $\mu = H$, and the logarithm $\ln(M_{\rm Pl}/H)$

arises from the running of $\Lambda_{\rm ren}(\mu)$ between the UV matching point $\mu = M_{\rm Pl}$ and the physical scale $\mu = H_0$. In the notation of the renormalisation group:

$$\Lambda_{\rm ren}(H) \;=\; \Lambda_{\rm ren}(M_{\rm Pl}) \;+\; \beta_\Lambda \ln\!\left(\frac{H}{M_{\rm Pl}}\right) \;+\; \cdots \tag{43}$$

The beta function β_Λ receives contributions from both the gravitational back-reaction (encoded in $V_{\rm grav}$) and the nonlocal infrared effects of the de Sitter horizon. On a quasi-de Sitter background, nonlocal terms of the schematic form $R\,\ln(\Box/\mu^2)\,R$ in the effective action evaluate to logs of H/μ, providing a physical (non-cutoff) origin for the logarithmic factor. In this reading, the back-reaction term (6) is an effective, integrated representation of these nonlocal contributions, and the scaling $\rho_* \sim H^2 M_{\rm Pl}^2/\ln(M_{\rm Pl}/H)$ is a statement about infrared running rather than a cutoff artifact.

We can extract β_Λ directly from the free-energy functional without recourse to QFT loop calculations.

Proposition 5.2 (Thermodynamic β-function)**.** *Define the thermodynamic response function*

$$\beta_\Lambda^{\rm thermo} \;:=\; \frac{\mathrm{d}}{\mathrm{d}\ln a}\,\frac{\partial\Phi}{\partial\rho}\bigg|_{\rho=\rho_*}. \tag{44}$$

Evaluating at the minimum ρ_* of Φ gives

$$\beta_\Lambda^{\rm thermo} \;=\; -\,\frac{H_0^2\,M_{\rm Pl}^2}{\big[\ln(M_{\rm Pl}/H)\big]^2}\;\cdot\;\frac{\mathrm{d}\ln H}{\mathrm{d}\ln a}, \tag{45}$$

Evaluating in the matter era where $\mathrm{d}\ln H/\mathrm{d}\ln a = -3/2$:

$$\beta_\Lambda^{\rm thermo}\bigg|_{\rm matter} \;=\; +\,\frac{3}{2}\,\frac{H_0^2\,M_{\rm Pl}^2}{\big[\ln(M_{\rm Pl}/H)\big]^2} \;>\; 0. \tag{46}$$

The positive sign means that $\Lambda_{\rm eff}$ increases as the universe expands and H decreases—consistent with the transition from deceleration to acceleration.

Proof. The renormalised stationarity condition (see Section 5.3) implicitly defines $\rho_* = \rho_*(H)$, since Φ depends on H through the de Sitter temperature $T_{\rm dS} = H/(2\pi)$ and the IR cutoff $\Lambda_{\rm IR} = H$. By the implicit function theorem, differentiating $\mathrm{d}\Phi/\mathrm{d}\rho\big|_{\rho_*} = 0$ with respect to $\ln a$ at fixed UV cutoff:

$$\frac{\partial^2\Phi}{\partial\rho^2}\bigg|_{\rho_*}\cdot\frac{\mathrm{d}\rho_*}{\mathrm{d}\ln a} \;=\; -\frac{\partial^2\Phi}{\partial\rho\,\partial H}\bigg|_{\rho_*}\cdot\frac{\mathrm{d}H}{\mathrm{d}\ln a}.$$

The dominant H-dependence enters through $B \propto T_{\rm dS} \propto H$. In the renormalised scaling $\rho_*^{\rm ren} \sim H^2 M_{\rm Pl}^2/\ln(M_{\rm Pl}/H)$, the logarithmic derivative $\mathrm{d}\ln\rho_*/\mathrm{d}\ln H \approx 2 + 1/\ln(M_{\rm Pl}/H) \approx 2$ gives $\mathrm{d}\rho_*/\mathrm{d}\ln a \approx 2\rho_*\,\mathrm{d}\ln H/\mathrm{d}\ln a$. The thermodynamic beta function (44) then evaluates to (45). In the matter era, $H \propto a^{-3/2}$, so $\mathrm{d}\ln H/\mathrm{d}\ln a = -3/2$, and the two minus signs (one from the formula, one from $\mathrm{d}\ln H/\mathrm{d}\ln a < 0$) combine to give $\beta_\Lambda^{\rm thermo} > 0$. $\square$

Remark 5.3. Proposition 5.2 is a *consistency check*, not an independent derivation. It presupposes the conjectured renormalised scaling $\rho_*^{\rm ren} \sim H^2 M_{\rm Pl}^2/\ln(M_{\rm Pl}/H)$ (cf. the conditional status of Theorem 3.1) and computes its logarithmic derivative—the resulting beta function is therefore a *tautological consequence* of the postulated scaling, not evidence for it. It does show that the sign and magnitude are internally consistent, but provides no independent support for the scaling itself. The sign and magnitude are consistent with the scalar-tensor Lagrangian (14): the scalaron-mediated running reproduces (45) at tree level.

Remark 5.4 (Semiclassical effective action). The one-loop effective action for $f(R) = R + \gamma R^2$ gravity takes the form

$$\Gamma_{1-\text{loop}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \gamma R^2 + \frac{1}{(4\pi)^2}\left(c_1 R^2 \log\frac{R}{\mu^2} + c_2 R_{\mu\nu}R^{\mu\nu}\log\frac{R}{\mu^2}\right)\right],$$

where c_1, c_2 are scheme-dependent constants and μ is the renormalisation scale. The non-local terms $R\log(\Box/\mu^2)R$ arise from the heat-kernel expansion and encode the running of γ with scale. Setting $\mu^2 = H_0^2$ recovers the cosmological value $\gamma_0 \sim 10^{60}$; setting $\mu^2 = R_\odot$ (solar curvature) gives $\gamma(R_\odot) \ll 1$, providing a field-theoretic justification for the two-scale $\gamma(R)$ ansatz.

Remark 5.5 (Beta function for Λ_{eff}). In the Wilsonian effective field theory framework, the cosmological constant runs with the RG scale k:

$$\beta_\Lambda = k\frac{d\Lambda(k)}{dk} = \frac{k^4}{16\pi^2}\left(\sum_i(-1)^{F_i}n_i - \frac{\Lambda(k)}{k^2}\right),$$

where the sum runs over particle species with spin-statistics factor $(-1)^{F_i}$ and multiplicity n_i. The FST functional $\Phi[\rho]$ provides the matching condition at the IR scale $k = H_0$: $\Lambda(H_0) = \Lambda_{\text{eff}} \sim H_0^2$. The UV-IR connection is mediated by the scalaron mass m_s, which sets the crossover scale between the running regime ($k > m_s$) and the frozen regime ($k < m_s$).

5.4 One-loop robustness

The scalar-tensor Lagrangian (14) contains the R^2 coupling γ, which introduces an additional scalar degree of freedom (the scalaron) with mass $m_s^2 = 1/(6\gamma)$. Its one-loop contribution to the vacuum energy density is of order

$$\delta\rho_s \sim \frac{m_s^4}{64\pi^2} = \frac{1}{64\pi^2\,(6\gamma)^2}. \tag{47}$$

For the tree-level minimum $\rho_* \sim 3M_{\text{Pl}}^2 H_0^2$ not to be overwhelmed, we require $\delta\rho_s \ll \rho_*$, which yields the parametric condition

$$\gamma \gg \frac{1}{M_{\text{Pl}}\,H_0} \sim 10^{60} \text{ in Planck units.} \tag{48}$$

For such large values of γ the scalaron becomes extremely light ($m_s \ll H_0$) and effectively decouples from late-time dynamics. Alternatively, one may interpret ρ_* as a renormalised matching datum, in which case $\delta\rho_s$ is absorbed into the renormalisation condition for Λ_{ren}; this is the standard viewpoint in quadratic gravity [14].

Standard Model fields couple to ϕ only through gravity (there are no direct Yukawa or portal couplings in the Lagrangian (14)). Consequently, matter-induced corrections to the scalar-field mass are Planck-suppressed: $\delta m_\phi^2 \sim m_{\text{heavy}}^2/(16\pi^2 M_{\text{Pl}}^2)$, which preserves the ultra-light quintessence mass $m_\phi \sim H_0$ without adding a separate scalar-mass fine-tune beyond the gravitational-sector assumptions. If, however, non-gravitational couplings were present, radiative corrections would generically generate $\delta m_\phi^2 \sim y^2 m_{\text{heavy}}^2/(16\pi^2)$, violating the technical naturalness of the Pöschl–Teller potential. The absence of such couplings is therefore a structural requirement of the model, consistent with the purely geometric origin of the scalar field in the R^2 sector.

5.5 Renormalisation-group interpretation of Φ

The free energy functional $\Phi[\rho]$ admits a natural interpretation as a Callan–Symanzik flow between UV and IR fixed points.

Definition 5.6 (RG flow of Φ)**.** Define the running free energy at scale μ:

$$\Phi(\rho;\,\mu) \;=\; \int d^3x \left[\rho \,\log \frac{\rho}{\rho_0(\mu)} - \rho + \rho_0(\mu) + \frac{G(\mu)}{2}\,\frac{\rho^2}{k_{\text{IR}}^2(\mu)}\right],$$

where $\rho_0(\mu)$, $G(\mu)$, and $k_{\text{IR}}(\mu)$ run with the RG scale μ. The Callan–Symanzik equation is

$$\left(\mu\,\frac{\partial}{\partial\mu} + \beta_G\,\frac{\partial}{\partial G} + \beta_k\,\frac{\partial}{\partial k_{\text{IR}}} + \gamma_\rho\,\rho\,\frac{\partial}{\partial\rho}\right)\Phi = 0.$$

Proposition 5.7 (Fixed-point structure)**.** *The RG flow has two fixed points:*

1. UV fixed point *($\mu \to M_{\text{Pl}}$): $\rho_0 \sim M_{\text{Pl}}^4/(8\pi G)$, $k_{\text{IR}} \sim M_{\text{Pl}}$. The free energy is dominated by the mode-counting term: $\Phi \sim M_{\text{Pl}}^4 \cdot \text{vol}(M)$. This is the cosmological constant problem in its raw form.*

2. IR fixed point *($\mu \to H_0$): $\rho^* = \rho_0(H_0) \sim H_0^2/(8\pi G)$, $k_{\text{IR}} \sim H_0$. The free energy is minimised at the de Sitter vacuum: $\Phi[\rho^*] = 0$. The effective cosmological constant $\Lambda_{\text{eff}} = 8\pi G\,\rho^*$ is determined by the IR scale.*

The flow from UV to IR is monotone: $\mu\,\frac{d}{d\mu}\,\Phi[\rho^*(\mu)] \;\leq\; 0$ (the free energy decreases under coarse-graining). This is a monotonicity property reminiscent of Zamolodchikov's c-theorem, though it follows from simpler statistical-mechanical arguments (the data-processing inequality for the KL divergence) rather than the 2D conformal field theory machinery of the original theorem.

Proof sketch. Monotonicity follows from the convexity of the KL divergence: at each RG step, the coarse-grained density $\rho_{k+1} = \mathcal{R}_k\,\rho_k$ satisfies $D_{\text{KL}}(\rho_{k+1}\|\rho_0) \leq D_{\text{KL}}(\rho_k\|\rho_0)$ by the data-processing inequality. The gravitational term $G\rho^2/k_{\text{IR}}^2$ is a relevant perturbation in the RG sense (dimension 2 in 4D), which grows towards the IR and stabilises the fixed point. □

Remark 5.8 (Connection to the resolvent architecture)*.* The RG flow of Φ exhibits the Pattern A resolvent structure:

- *Leading neutral:* At the UV fixed point, $\Phi \sim M_{\text{Pl}}^4$ is scale-independent (zeroth order).

- *First-order degenerate:* The one-loop correction $\delta\Phi \sim \log(\mu/M_{\text{Pl}})$ is a logarithmic correction that does not break the UV fixed point.

- *Second-order dominant:* The gravitational self-interaction $G\rho^2/k_{\mathrm{IR}}^2$ is the relevant perturbation that drives the flow to the IR fixed point and determines Λ_{eff}.

This is structurally identical to the resolvent hierarchy in the Yang–Mills (Poincaré constant under RG) and Navier–Stokes (Gronwall constant under Galerkin truncation) companion papers.

Theorem 5.9 (Cosmological RFEP fixed-point universality – conditional)**.** *Let $\Phi[\rho;\,\mu]$ be a running free-energy functional satisfying:*

(C1) IR cutoff by causal structure: *the infrared scale $k_{\mathrm{IR}}(\mu)$ is bounded below by the Hubble parameter $H(\mu)$;*

(C2) Positive gravitational feedback: *the self-interaction term satisfies $V[\rho] \geq c\, G\rho^2/k_{\mathrm{IR}}^2$ for some $c > 0$;*

(C3) Entropic regularisation: *Φ contains a KL-divergence term $D_{\mathrm{KL}}(\rho\|\rho_0)$ ensuring strict convexity.*

Then the RG flow $\mu \mapsto \Phi[\rho^*(\mu);\,\mu]$ possesses a non-zero IR fixed point with

$$\Lambda_{\mathrm{eff}} \;\sim\; \frac{H_0^2}{\ln(M_{\mathrm{Pl}}/H_0)} \;\neq\; 0.$$

In particular, conditions (C1)–(C3) *exclude $\Lambda_{\mathrm{eff}} = 0$: a vanishing cosmological constant is thermodynamically unstable.*

Proof sketch. By (C3), Φ is strictly convex, so $\rho^*(\mu)$ exists and is unique at each scale. By (C2), the gravitational term is a relevant perturbation (dimension 2 in 4D) that grows under coarse-graining $\mu \to 0$. By (C1), the flow cannot proceed below $\mu = H_0$, fixing the IR endpoint. The monotonicity $\mu \frac{d}{d\mu}\Phi \leq 0$ (data-processing inequality for KL divergence, cf. Proposition 5.7) ensures convergence to a fixed point. At the fixed point, the balance between entropic penalty and gravitational feedback gives $\rho^* \sim H_0^2/(8\pi G \ln(M_{\mathrm{Pl}}/H_0))$, hence $\Lambda_{\mathrm{eff}} = 8\pi G\rho^* \neq 0$.

Note: The scaling $\rho^* \sim H_0^2/(8\pi G\,\ln(M_{\mathrm{Pl}}/H_0))$ relies on the same renormalisation-group matching argument as Theorem 3.1 and carries the same "conditional" status: it is conjectured, not derived from the raw mode sum alone. □

Remark 5.10 (UV-completion independence)*.* Conditions (C1)–(C3) are imposed at the *IR* end of the RG flow. A UV completion—string landscape, asymptotic safety à la Wetterich [30], loop quantum gravity—would have to reproduce an IR flow satisfying these conditions for the present mechanism to apply. This suggests a possible universality class behind $\Lambda_{\mathrm{eff}} \sim H_0^2/\ln(M_{\mathrm{Pl}}/H_0)$, but it is not derived here as a theorem about all UV-complete quantum-gravity theories. In this paper, the cosmological constant problem is reduced to an IR-selection problem once the RG-matching assumption is granted; the UV matching itself remains an explicit open gate.

Remark 5.11 (RG time as cosmic time)*.* The identification $\mu \leftrightarrow H(t)$ maps the RG flow to cosmological evolution: coarse-graining (μ decreasing) corresponds to expansion (H decreasing). The Friedmann equation $H^2 = 8\pi G\rho/3$ then plays the dual role of an equation of motion *and* an RG flow equation, with $\Lambda_{\mathrm{eff}}(t) \sim H(t)^2/\ln(M_{\mathrm{Pl}}/H(t))$ tracking the Hubble rate adiabatically. This dissolves the coincidence problem: Λ_{eff} and H^2 are not accidentally comparable—they are locked together by the RG flow. The beta function $\beta_\Lambda = \mu\,\partial\Lambda/\partial\mu$ is the Friedmann deceleration parameter in disguise.

5.6 Predictions: Slow evolution of Λ_{eff}

If the effective cosmological constant evolves as $\Lambda_{\text{eff}} \sim H^2/\ln(M_{\text{Pl}}/H)$, then Λ_{eff} is *not* exactly constant but decreases slowly as the universe expands and H decreases. The resulting intrinsic dark-energy equation-of-state parameter satisfies $w_{\text{DE}} = -1 + \varepsilon$ with

$$\varepsilon \;\sim\; \frac{2}{\ln(M_{\text{Pl}}/H_0)} \;\approx\; 0.014, \tag{49}$$

which is consistent with current constraints ($w = -1.03 \pm 0.03$, Planck 2018) and with the DESI DR2 results [10], which show 2–4σ evidence for dynamical dark energy ($w \neq -1$). Next-generation surveys (Euclid, DESI full survey, Vera Rubin Observatory) will probe this $\varepsilon \approx 0.014$ deviation at the required precision.

5.7 Thermodynamic selection of the vacuum energy

The variational result of Theorem 3.1 admits a complementary thermodynamic interpretation through the Maximum Entropy Production Principle (MEPP) [8]. Among all kinematically admissible vacuum energy densities, the realised value ρ_* may be interpreted as the one that maximises the rate of entropy production $\dot{S}$ over cosmological time scales. This selection criterion is consistent with our variational bound: the minimum of the free-energy functional $\Phi(\rho)$ may be identified with the maximum of $\dot{S}$, because the entropy term in Φ acts as a thermodynamic penalty that suppresses both excessively large and vanishingly small vacuum energies. At the saddle-point, the de Sitter horizon radiates at temperature $T_{\text{dS}} = H/(2\pi)$, and the associated Gibbons–Hawking entropy $S_{\text{dS}} = \pi c^5/(G\hbar H^2)$ reaches a quasi-stationary maximum—the universe expands just fast enough to maximise the accessible phase-space volume.

From this perspective, the accelerated expansion is not a coincidence but a thermodynamic imperative. Once stellar nucleosynthesis exhausts the available baryonic free energy, the dominant channel for entropy production becomes the cosmological expansion itself. The vacuum energy density $\rho_* \sim H_0^2/G$ is precisely the value at which the horizon entropy production rate is maximised, subject to the constraint that ρ must be sustained self-consistently by the gravitational back-reaction term (6). This thermodynamic selection mechanism offers a physical reason *why* the cosmological constant takes the value it does, complementing the variational *how* provided by the free-energy functional.

In this reading, the coincidence problem dissolves entirely: the vacuum energy tracks the matter density not by accident but because both are governed by the same entropy-maximising dynamics. The infrared cutoff $\Lambda_{\text{IR}} = H$ couples the vacuum sector to the expansion history, ensuring that ρ_* adjusts adiabatically as the universe evolves from matter domination to dark-energy domination.

Remark 5.12 (2D parameter scan as quantitative coincidence analogue). The two-dimensional parameter scan over (α, γ) performed in the companion quantitative study [20] provides a concrete realisation of the coincidence-problem resolution. The free-energy functional $\Phi[\rho]$ exhibits a *sharp minimum* in the (α, γ)-plane, automatically selecting the observed order of magnitude of the effective cosmological constant Λ_{eff}. This sharpness is not imposed by fine-tuning but emerges from the interplay of the entropy penalty (logarithmic, favouring small ρ) and the gravitational back-reaction (quadratic, penalising vanishing ρ). The 2D scan thus elevates the coincidence-problem resolution from a qualitative argument ("Λ_{eff} tracks H^2") to a quantitative demonstration: the functional landscape itself forces ρ_* into a narrow valley whose depth is set by H_0^2/G.

Proposition 5.13 (Thermodynamic selection of Λ_{eff} – conditional on (C2))**.** *Let $\Phi[\rho]$ be a free energy functional on cosmological density profiles satisfying:*

(C1) General covariance: *Φ is invariant under coordinate transformations.*

(C2) Minkowski subtraction: *The bare cosmological constant $\Lambda_0 \sim M_{\text{Pl}}^4$ is absorbed by renormalisation: $\Phi[\rho] = \Phi_{\text{ren}}[\rho - \rho_0]$ where ρ_0 is the Minkowski vacuum.* ***Warning:*** *This condition assumes that a consistent renormalisation scheme exists in which the Minkowski vacuum has $\Lambda = 0$. In the Standard Model, this is a* physical input *(renormalisation condition), not a derived result. Condition (C2) therefore* presupposes *a solution to the classical cosmological constant problem at the level of UV subtraction.*

(C3) Stability: *Φ has a unique minimiser ρ^* with $\nabla^2\Phi[\rho^*] > 0$.*

Then the effective cosmological constant satisfies $\Lambda_{\text{eff}} = 8\pi G\rho^* \sim H_0^2$, determined by the infrared scale H_0 rather than the ultraviolet scale M_{Pl}. The proposition is conditional: it shows that if *the UV problem (C2) is resolved, then the IR dynamics uniquely select the observed order of magnitude. It does not solve the UV problem itself. Note that the logical content is modest: given (C2), the scaling $\Lambda_{\text{eff}} \sim H_0^2$ follows essentially from dimensional analysis in the renormalised theory plus (C3). The non-trivial open question is establishing (C2) itself.*

Proof sketch. (C1) constrains Φ to depend only on scalar invariants of ρ. (C2) removes the M_{Pl}^4 contribution, leaving only IR-scale terms. (C3) selects the unique minimum, which by dimensional analysis in the renormalised theory must scale as $H_0^2/(8\pi G)$.

Caveat regarding (C2). Condition (C2) assumes that the bare cosmological constant $\Lambda_0 \sim M_{\text{Pl}}^4$ "is absorbed by renormalisation"—but this absorption is precisely the step that constitutes the cosmological constant problem in its classical form. In the Standard Model, the Minkowski vacuum does *not* have $\Lambda = 0$; that assignment is a *physical renormalisation condition*, not a derived result. Condition (C2) therefore encodes the assumption that a consistent renormalisation scheme exists in which the Minkowski vacuum has vanishing cosmological constant. This is a necessary input, not a consequence of the thermodynamic selection mechanism. The theorem should be read as: *given* that (C2) can be imposed consistently, the IR dynamics select $\Lambda_{\text{eff}} \sim H_0^2$. A proof that (C2) is self-consistent (i.e., that no obstruction prevents setting $\Lambda_{\text{Mink}} = 0$) remains an open problem. □

5.8 Falsifiable predictions

The scalar-tensor framework of Section 3.1 generates a set of quantitative predictions that distinguish it from both ΛCDM and generic modified-gravity models. We list the principal observational signatures:

1. **Gravitational slip parameter.** The scalar-tensor Lagrangian (14) modifies the two gravitational potentials Ψ and Φ_{N} in the conformal Newtonian gauge $\mathrm{d}s^2 = -(1+2\Psi)\,\mathrm{d}t^2 + a^2(1+2\Phi_{\text{N}})\,\mathrm{d}\mathbf{x}^2$. In the quasi-static, sub-horizon approximation, the modified Poisson equations for the $R + \gamma R^2$ sector read [13, 14]:

$$k^2\,\Phi_{\text{N}} = -4\pi G\,a^2\,\bar{\rho}\,\delta\left[1 + \frac{1}{3}\,\frac{k^2/a^2}{k^2/a^2 + m_s^2}\right], \tag{50}$$

$$k^2\,\Psi = -4\pi G\,a^2\,\bar{\rho}\,\delta\left[1 + \frac{2}{3}\,\frac{k^2/a^2}{k^2/a^2 + m_s^2}\right], \tag{51}$$

where $m_s^2 = 1/(6\gamma)$ is the scalaron mass squared. The gravitational slip parameter follows immediately:

$$\eta(k) \;\equiv\; \frac{\Psi}{\Phi_{\rm N}} \;=\; \frac{1+\dfrac{2}{3}\,\dfrac{k^2/a^2}{k^2/a^2+m_s^2}}{1+\dfrac{1}{3}\,\dfrac{k^2/a^2}{k^2/a^2+m_s^2}} \;\xrightarrow{k\gg a\,m_s}\; \frac{1+\frac{2}{3}}{1+\frac{1}{3}} \;=\; \frac{5/3}{4/3} \;=\; \frac{5}{4}. \tag{52}$$

In ΛCDM, $\eta = 1$ exactly. The crossover occurs at $k_{\rm cross} \approx 0.86\,a\,m_s$, where $\eta = 1.125$. For a scalaron mass $m_s \sim H_0$ (as suggested by the Pöschl–Teller dynamics), the Compton wavelength $\lambda_C \sim c/H_0$ lies near the Hubble radius, so that Euclid's probing scales ($k \sim 0.01$–$0.2\,h\,{\rm Mpc}^{-1}$) are deep in the sub-Compton regime where $\eta \approx 5/4$. This 25% deviation from ΛCDM is within the sensitivity of Euclid's weak-lensing and galaxy-clustering measurements [13]. Crucially, purely geometric alternatives such as Finsler gravity [17] predict $\eta \neq 5/4$, making the gravitational slip a *model-discriminating observable* between the CRM, ΛCDM, and Finsler approaches.

2. **Lensing parameter.** The effective lensing parameter, defined as $\Sigma(k) \equiv (\Phi_{\rm N} + \Psi)/(2\Phi_{\rm N})$, evaluates to

$$\Sigma(k) \;=\; \frac{2+f(k)}{2+\frac{2}{3}\,f(k)}, \qquad f(k) \;:=\; \frac{k^2/a^2}{k^2/a^2+m_s^2}. \tag{53}$$

In the GR limit ($f \to 0$, i.e., $k \ll a\,m_s$): $\Sigma \to 1$. In the sub-Compton regime ($f \to 1$, i.e., $k \gg a\,m_s$): $\Sigma \to 9/8 = 1.125$. The scale-dependent deviation from $\Sigma = 1$ is a further distinguishing signature compared to ΛCDM (where $\Sigma = 1$ exactly) and is testable by weak-lensing surveys such as Euclid [14].

3. **Scalar-field oscillations in $H(a)$.** At late times ($a \gtrsim 2$), the scalar field ϕ oscillates around the minimum of $V_{\rm PT}$ with frequency $\omega_\phi = \sqrt{2V_0}/\phi_0$. These oscillations imprint a characteristic modulation on the Hubble parameter:

$$\frac{\delta H}{H} \;\sim\; \frac{\Phi_0}{2}\,\mathrm{sech}^2\!\Big[\kappa(a-a_{\rm trans})\Big]\;\cos(\omega_\phi\,t), \tag{54}$$

with an amplitude of order 10^{-3}–10^{-2} at $a \approx 2$. Future precision measurements of $H(z)$ from gravitational-wave standard sirens could detect this oscillatory signature.

4. **Connection to the MOND acceleration scale.** The characteristic acceleration scale of the scalar field is

$$a_0 \;=\; \frac{c\,H_0}{2\pi} \;\approx\; 1.0\times10^{-10}\ {\rm m\,s^{-2}}, \tag{55}$$

which coincides with the empirical MOND acceleration $a_0^{\rm MOND} \approx 1.2\times10^{-10}\ {\rm m\,s^{-2}}$ to within 20%. This numerical coincidence is *suggestive but not derived from the model*: the relation $a_0 \sim cH_0$ is a generic dimensional consequence of any cosmological acceleration scale and does not, by itself, establish a causal connection between dark energy and MOND dynamics. The factor 2π from the de Sitter temperature $T_{\rm dS} = \hbar\,a_0/(2\pi c\,k_{\rm B})$ narrows the agreement but a physical mechanism linking the Pöschl–Teller scalar field to galactic-scale modified dynamics remains to be established.

5. **Resolvent-vacuum prediction for η.** The second-order resolvent framework (Section 4.6) provides an independent derivation of $\eta = 5/4$: the balance between the entropic term (de Sitter temperature) and the gravitational back-reaction in $\Phi(\rho)$ fixes the ratio of the modified Poisson potentials at sub-Compton scales, yielding $\eta = 5/4$ as a necessary consequence of the resolvent structure rather than a parameter fit. Crucially, Finsler gravity models [17] predict $\eta \neq 5/4$, since their purely geometric modification of the Friedmann equations operates through a different mechanism (anisotropic spacetime metric rather than scalar-tensor coupling). This makes η a sharp *model-discriminating observable*:

   - **CRM/Resolvent:** $\eta = 5/4$ at $k \gg a\, m_s$.
   - **Finsler:** $\eta \neq 5/4$ (generically $\eta > 1$ but not 5/4).
   - **ΛCDM:** $\eta = 1$ exactly.

   The DESI full survey (expected 2028) and Euclid DR1 (expected 2027) will measure η at $z < 0.5$ with $\sim 2\%$ precision. A 2σ deviation from 5/4 would falsify the second-order resolvent framework for dark energy; convergence toward 5/4 would simultaneously rule out both ΛCDM and Finsler alternatives.

Each of these predictions is falsifiable with data expected within the next decade (Euclid DR1: 2027; LISA: 2030s; DESI full survey: 2028). A single confirmed deviation from ΛCDM along any of these channels would provide evidence for the scalar-tensor mechanism; a confirmed $\eta = 1$ at sub-Compton scales would rule it out.

Remark 5.14 (Hu–Sawicki model as FST instantiation). The Hu–Sawicki model [13]

$$f(R) = R - \mu^2 \frac{c_1 (R/\mu^2)^n}{1 + c_2 (R/\mu^2)^n},$$
$$\mu^2 = H_0^2 \Omega_m, \qquad \frac{c_1}{c_2} = \frac{6\Omega_\Lambda}{\Omega_m}.$$

provides a concrete realisation of the FST dark energy programme. The model parameters map to FST quantities as follows:

Hu–Sawicki	**FST functional $\Phi[\rho]$**
$f_{R0} = -n\, c_1/c_2^2$ $\times (H_0^2 \Omega_m / R_0)^{n+1}$	Coupling strength γ_{eff}
Scalaron mass $m_s^2 = (3 f_{RR})^{-1}$	Curvature of Φ at minimum
Compton wavelength $\lambda_C = 2\pi/m_s$	Crossover scale k_{cross}
n (steepness parameter)	Rate of chameleon screening

For $n = 1$ and $|f_{R0}| = 10^{-6}$, the scalaron mass at cosmological density is $m_s \sim 10^{-33}$ eV ($\lambda_C \sim 30$ Mpc), while at solar-system density $m_s \sim 10^{-3}$ eV ($\lambda_C \sim 0.2$ mm), providing efficient chameleon screening.

Unified $\eta = 5/4$ prediction. The quasi-static modified Poisson equations (50)–(51) with coefficients 1/3 (for Φ_N) and 2/3 (for Ψ) are *universal* for all metric $f(R)$ theories—they follow from the linearised trace equation and are independent of the specific functional form of $f(R)$ (cf. Hu & Sawicki 2007, eq. 12–13; Sotiriou & Faraoni 2010, Sec. V.B). The Hu–Sawicki model differs from the quadratic $R + \gamma R^2$ theory only in the *density dependence* of the scalaron mass $m_s^2(\rho)$, not in the Poisson-equation coefficients. In the sub-Compton limit ($k \gg a\, m_s$), both yield $\eta = (5/3)/(4/3) = 5/4$. The crucial advantage of Hu–Sawicki is that $m_s(\rho)$ grows with ambient density, providing chameleon screening in the solar system while preserving $\eta = 5/4$ on cosmological sub-Compton scales.

Regime guardrail. The value $\eta = 5/4$ is a quasi-static, linear-perturbation, unscreened sub-Compton statement: $k/a \gg m_s(z, \rho)$, outside the screened high-density regime and outside the super-Compton GR-recovery regime. It is not a solar-system prediction and not a background-level dark-energy fit parameter.

Erratum (v1). An earlier version of this remark stated that the Hu–Sawicki model yields $\eta \to 4/3$ due to different Poisson-equation coefficients. This was incorrect: the coefficients 1/3 and 2/3 are universal for all $f(R)$ theories. The coefficient ratio giving $\eta = 5/4$ applies uniformly across the linearised unscreened sub-Compton $f(R)$ regime, including Hu–Sawicki.

5.9 Exploratory MCMC Scan for Hu–Sawicki Parameters

We report an exploratory Markov chain Monte Carlo (MCMC) likelihood scan of the Hu–Sawicki $f(R)$ parametrisation, not a production cosmological parameter inference. The scan combines simplified diagonal Planck 2018 (TT/TE/EE) and DESI DR2 (BAO) terms with a Cassini hard prior on $|\gamma_{\rm PPN} - 1|$. The sampler employs 32 walkers over 3 000 steps with a 500-step burn-in, yielding 80 000 post-burn-in samples. These samples should be read as posterior-like stress-test samples, not as independent constraints from the official Planck/DESI likelihoods. Table 1 summarises the sampled parameters and two derived quantities.

The best-fit $\chi^2 = 6811.4$ for a combined data set with $N_{\rm data} \approx 7200$ effective data points (Planck TT/TE/EE ~ 7000 + DESI BAO ~ 12 + Cassini 1) gives $\chi^2/N_{\rm data} \approx 0.95$, indicating an acceptable fit. The modified-gravity amplitude $\alpha_{M,0} = 1.5 \times 10^{-5}\ (^{+2.0\times10^{-5}}_{-1.0\times10^{-5}})$ is consistent with zero at $< 2\sigma$. This means that the data *do not require* a departure from GR; the GR limit ($\alpha_{M,0} = 0$) lies within the 2σ posterior. The honest interpretation is that current data are compatible with both GR and the mild FST modifications, but do not favour FST over ΛCDM. The steepness parameter $n_{\rm exp} \approx 0.71$ favours moderate nonlinearity, broadly consistent with the $n = 1$ Hu–Sawicki benchmark. The standard cosmological parameters—cold dark matter density $\omega_{\rm cdm}$, primordial amplitude $\ln(10^{10}A_s)$, and spectral index n_s—are all consistent with the Planck 2018 best-fit values, confirming that the Hu–Sawicki extension does not distort the well-constrained ΛCDM sector.

Methodological caveats. (i) The likelihood is deliberately simplified: it uses diagonal CMB and BAO summary terms rather than the official Planck and DESI likelihood pipelines, and it treats the Cassini condition as a hard prior rather than a full screened-scalar calculation. (ii) Chain convergence has not been verified by Gelman–Rubin ($\hat{R}$) or Geweke diagnostics; the 500-step burn-in and 3 000-step chain length should be treated as preliminary. A production-quality analysis would require $\hat{R} < 1.01$ for all parameters and an effective sample size $n_{\rm eff} > 1000$. (iii) The posterior $\alpha_{M,0} \approx 0$ is equally consistent with GR ($\alpha_{M,0} = 0$) and with the FST prediction ($\alpha_{M,0} > 0$); the data do not currently discriminate between the two. A Bayesian model comparison (Bayes factor ΛCDM vs. Hu–Sawicki) would quantify this more precisely but is beyond the scope of this skeleton draft. (iv) The best-fit $|f_{R0}| = 1.2 \times 10^{-5}$ is above the commonly cited cosmological upper bound $|f_{R0}| < 6 \times 10^{-5}$ (Cataneo et al. 2015, 95% CL from cluster abundances) but may be in tension with the Cassini-derived bound $|f_{R0}| \lesssim 10^{-6}$ in the linearised regime. The compatibility of our best-fit value with solar-system constraints relies on the nonlinear chameleon mechanism at this specific f_{R0}, which has not been explicitly verified here. (v) Prior sensitivity: the flat prior $\alpha_{M,0} \in [0, 0.01]$ includes GR ($\alpha_{M,0} = 0$) at the boundary; a prior excluding zero would produce a qualitatively different posterior.

Remark 5.15 (ΛCDM comparison). The best-fit $\chi^2 = 6811.4$ for the Hu–Sawicki model should be compared with the ΛCDM baseline. For the same data combination (Planck 2018 TT/TE/EE + DESI DR2 BAO), ΛCDM achieves $\chi^2 \approx 6810$–6815 (depending on the exact likelihood implementation). The Hu–Sawicki model with its additional parameters ($\alpha_{M,0}$, $n_{\rm exp}$) therefore does *not* improve the fit—the extra parameters are unconstrained, consistent with GR ($\alpha_{M,0} = 0$ within 2σ). This is explicitly acknowledged: the data do not currently require modified gravity.

Table 1: Exploratory Hu–Sawicki MCMC scan from simplified Planck 2018 + DESI DR2 + Cassini terms (80 000 post-burn-in samples, 32 walkers, 3 000 steps).

Parameter	Median	$+1\sigma$	-1σ	Prior
$\alpha_{M,0}$	1.5×10^{-5}	$+2.0 \times 10^{-5}$	-1.0×10^{-5}	$[0,\ 0.01]$
$n_{\rm exp}$	0.711	+0.576	−0.319	$[0.1,\ 4]$
$\omega_{\rm cdm}$	0.12005	+0.00031	−0.00030	$[0.10,\ 0.14]$
$\ln(10^{10} A_s)$	3.0447	+0.0020	−0.0019	$[2.9,\ 3.2]$
n_s	0.9654	+0.0024	−0.0025	$[0.9,\ 1.0]$
$\lvert f_{R0}\rvert$	$1.2 \times 10^{-5}\ \binom{+7.9\times10^{-6}}{-7.9\times10^{-6}}$			derived
$\lvert\gamma_{\rm PPN} - 1\rvert$ proxy	$1.2 \times 10^{-5} < 2.3 \times 10^{-5}$			exploratory prior

The derived scalar-field amplitude $|f_{R0}| = 1.2 \times 10^{-5}$ passes only the linear proxy $|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$ used in this exploratory likelihood. It does *not* close the solar-system viability question, because Lemma 4.14 uses the conservative guardrail $|f_{R0}| \leq 10^{-6}$ and the chameleon thin-shell condition has not been explicitly verified for this point. The MCMC scan is therefore evidence for background-level compatibility, not a proof of Cassini-safe screening.

The small but non-zero $|f_{R0}|$ leaves room for the regime-limited FST target $\eta = 5/4$ on unscreened quasi-static sub-Compton scales (cf. Remark 5.14), while the solar-system screening gate remains open until the thin-shell calculation is supplied. These results show only that the simplified Planck+DESI+Cassini stress test is compatible with the FST/$f(R)$ framework and does not exclude mild modifications, but it does not prefer them over ΛCDM.

5.10 Competing approaches and recent developments

Several independent lines of research corroborate or challenge the thermodynamic derivation of the cosmological constant presented here.

Holographic entropy and MOND. Sevinç, Ökcü, and Aydiner [16] derive modified Friedmann equations from generalised entropy–area relations combined with the Extended Uncertainty Principle (EUP), establishing a direct mathematical link between the MOND acceleration scale and holographic entropy bounds. This independently confirms that holographic thermodynamic arguments can modify the macroscopic gravitational dynamics in a manner consistent with our equation (55).

Finsler gravity. Pfeifer et al. [17] demonstrate that kinematic gases propagating in a Finsler (non-Riemannian) spacetime naturally produce modified Friedmann equations predicting accelerated expansion without any dark energy component or scalar field. This purely geometric explanation constitutes a strong methodological competitor to the scalar-tensor mechanism of the CRM: it achieves the same phenomenology through spacetime geometry alone, without thermodynamic or dissipative selection principles. A key discriminant between the two approaches is the gravitational slip parameter η: the CRM predicts $\eta = 5/4$ at sub-Compton scales, whereas pure Finsler modifications generically predict $\eta \neq 5/4$.

Informational persistence. As a non-peer-reviewed Zenodo preprint, Hunt [18] interprets dark-matter phenomenology as the survival of curvature information under coarse-graining. This is useful only as a low-weight conceptual comparator for information-theoretic modified-gravity language, not as empirical support for the CRM.

5.11 Post-Zookeeper transfer: DESI DR2, JWST and TD-COSMO as a data overlay

The post-Zookeeper role of the present framework is to make the CRM observable rather than merely structural. The next version should add a data-overlay layer that maps CRM parameters directly to the standard late-time descriptors used by current surveys.

Relative to the current CORE status, the Zookeeper closure of Riemann-$C2/MS2$ is not a cosmology theorem. CRM inherits only the RFEP v1.8 adapter normal form: operator/form, defect, approximation, gap, and limiting passage must be made explicit in the Dark-Energy domain itself. The open work therefore remains the CRM-to-CPL map, the joint DESI/JWST/TDCOSMO overlay, and the Cassini/BBN screening guardrails.

The transfer slots are:

- **Operator/form.** The Flow-Zeta or RG operator for $w(z)$, $H(z)$, and Λ_{eff}.
- **Defect.** The joint residual against DESI BAO, CMB, local Cepheid/SN measurements, strong-lensing time delays, UV matching, and screening constraints.
- **Approximation.** CPL (w_0, w_a), Hu–Sawicki parameter grids, MCMC samples, and redshift-bin likelihoods.
- **Gap.** RFEP fixed-point curvature, de Sitter stability, and the screening viability margin.
- **Limit.** Background-only fits $\to$ perturbations/CMB/BBN consistency $\to$ local-gravity tests such as Cassini.

In Zookeeper/RFEP language this is a residual-over-gap programme. The paper already proves the convex/free-energy selection of the vacuum; a future data version should additionally display the estimate

$$\|(I - P_{\text{CRM}})\Theta_{\text{data}}\| \lesssim \frac{r_{\text{DESI}} + r_{\text{local}} + r_{\text{screen}}}{g_{\text{dS}}},$$

where the numerator collects survey, local-distance and screening residuals, while g_{dS} denotes the curvature of the selected de Sitter/free-energy branch. This is a bookkeeping target, not a new claim already established in the present version.

Concretely, the next technical target is a CRM-to-CPL map

$$\Theta_{\rm CRM} \longmapsto (w_0, w_a, H_0^{\rm eff}, \Omega_m, \Omega_{\rm DE}),$$

followed by a comparison table for the main DESI DR2 combinations:

- DESI+CMB
- DESI+CMB+Pantheon+
- DESI+CMB+Union3
- DESI+CMB+DESY5

The same table should also record whether the parameter point remains compatible with JWST/Cepheid local-distance results, TDCOSMO strong-lensing constraints, BBN protection of the sound horizon, and the Cassini bound on post-Newtonian deviations.

This reframes the claim. The model should no longer merely state that evolving dark energy is qualitatively compatible with CRM. It should identify a CRM parameter window that simultaneously fits the DESI DR2 preference for dynamical dark energy, stays compatible with the local-H_0 triangle, and does not violate BBN or Solar-System screening.

5.12 Open directions

Remark 5.16 (Cosmological phase-transition chain)*.* The cosmological history can be read as a sequence of Nash phase transitions in the functional $\Phi[\rho]$: radiation domination $\to$ matter domination $\to$ dark energy domination, each corresponding to a change in the dominant term of the free energy.

Remark 5.17 (Legendre–Fenchel duality for dark energy)*.* The Legendre–Fenchel duality $S \geq F$ provides an independent consistency check: the thermodynamic entropy of the de Sitter horizon must exceed the free energy of the vacuum configuration, constraining $\Lambda_{\rm eff}$ from above.

Remark 5.18 (England inequality for expansion rate)*.* An analogue of England's dissipative adaptation inequality may constrain the cosmic expansion rate: $\dot{a}/a \geq \Delta S/(k_B\,\Delta t)$, linking the Hubble parameter to the entropy production rate at the cosmological horizon.

Remark 5.19 (Topological charge as IR constraint)*.* The functional $\Phi[\rho]$ can be extended to $\Phi[\rho, Q]$ where Q is a topological charge (Hopf invariant) characterising the field configuration. Minimisation at fixed Q yields sector-dependent vacuum energies $\Lambda_{\rm eff}(Q)$, with the physical vacuum corresponding to $Q = 0$ (trivial topology). Non-trivial sectors ($Q \neq 0$) contribute to the partition function with an entropy suppression $e^{-c|Q|^{3/4}}$ (Faddeev–Skyrme bound [29]), providing a natural hierarchy between the cosmological vacuum and topological defects. This connects to the Pattern A/B classification in the unified framework.

Remark 5.20 (Hopfion energy bounds as backreaction)*.* The Faddeev–Skyrme energy bound $E \geq c|Q|^{3/4}$ for knotted solitons [29] provides a non-quadratic alternative to the gravitational self-interaction $V_{\rm grav} \sim G\rho^2/k^2$. If the dark energy vacuum supports topological excitations (knotted scalar field configurations), the backreaction cost function acquires a topological contribution: $V_{\rm eff}[\rho] = V_{\rm grav}[\rho] + \lambda|Q[\rho]|^{3/4}$. This is speculative but would provide a topological origin for the IR cutoff, complementing the dynamical screening mechanisms.

Remark 5.21 (MOND from generalised entropy-area relations). Sevinç, Ökcü, and Aydiner [16] derive modified Friedmann equations from generalised entropy-area relations (Extended Uncertainty Principle). Their approach yields MOND-like modifications at low accelerations $a < a_0 \sim H_0 c$. The FST functional $\Phi[\rho]$ shares the thermodynamic starting point but differs in mechanism: Φ minimises free energy with stability constraints, while the entropy-area approach modifies the Bekenstein–Hawking entropy directly. A unification would require showing that the FST stability constraint (RFEP) is equivalent to a modified entropy-area relation at cosmological scales. This remains an open question.

Remark 5.22 (Relation to Dynamic Vacuum Field Theory). Dynamic Vacuum Field Theory (DVFT) models the vacuum as a dynamical scalar field and derives MOND-like phenomenology from vacuum fluctuations. The key similarities and differences with the FST approach:

- *Shared:* Both use a scalar field to mediate dark energy effects and predict deviations from ΛCDM at galactic scales.
- *Different:* DVFT lacks a thermodynamic/entropy argument; the FST functional $\Phi[\rho]$ is derived from the RFEP (a variational principle), not postulated. DVFT does not predict a specific value of η; FST targets $\eta = 5/4$ on quasi-static, linear, unscreened sub-Compton metric-$f(R)$ scales.

The observational discriminant is clear only within that regime: if $\eta \approx 1.25$ is confirmed by Euclid after scale cuts and screening systematics are controlled, DVFT (which does not predict $\eta > 1$ generically) would be disfavoured relative to FST.

Remark 5.23 (Holographic bulk-boundary matching and density-correlated constraints). Two related extensions of the RG framework:

1. *Holographic matching:* In an AdS/CFT-inspired picture, the UV divergence $\Lambda_0 \sim M_{\rm Pl}^4$ is a bulk quantity, while $\Lambda_{\rm eff} \sim H_0^2$ is a boundary (horizon) quantity. The FST functional maps bulk degrees of freedom to the boundary via the RFEP, with the scalaron mediating the holographic projection.
2. *Density-correlated constraint:* The scalaron mass m_s can couple non-linearly to the trace $T^\mu{}_\mu$ of the stress-energy tensor: $m_s^2(x) = m_{s,0}^2(1 + \alpha |T^\mu{}_\mu| / M_{\rm Pl}^2)$. In dense environments ($|T^\mu{}_\mu| \gg M_{\rm Pl}^2 H_0^2$), m_s diverges and the scalaron decouples, providing yet another screening route complementary to chameleon and Vainshtein.

Remark 5.24 (Relation to asymptotic safety). Wetterich [30] derives dark energy evolution from an asymptotically safe UV fixed point in quantum gravity. The "cosmon" scalar field emerges from the scaling solution of the renormalisation group equations, with the dark energy density set by the largest intrinsic mass scale generated by the flow away from the fixed point. The FST approach shares the RG perspective but differs structurally: whereas asymptotic safety determines Λ from the UV fixed-point structure, the FST functional selects $\Lambda_{\rm eff}$ via IR thermodynamic equilibrium. Both predict slow evolution of Λ, making them observationally compatible but theoretically distinct. State-of-the-art truncation results [12, 11] provide growing numerical evidence that the Reuter fixed point constrains Standard Model parameters (top-quark mass, Higgs mass, neutrino masses), lending empirical support to the fixed-point structure that the FST framework inherits as its UV completion.

Remark 5.25 (Relation to Covariant Canonical Gauge Gravity). The CCGG framework of Vasak, Struckmeier, and Kirsch [31] derives dark energy and inflation from locally contorted spacetime within a De Donder–Weyl Hamiltonian formulation. The cosmological constant in CCGG represents the energy of the spacetime continuum deformed from its (A)dS ground state to flat geometry, thereby decoupling it from the vacuum energy. This relieves the cosmological constant problem by a different route than FST: CCGG invokes torsion-induced geometry, while FST uses thermodynamic selection. Both approaches predict a time-dependent effective Λ; their observational signatures differ in the gravitational slip (a regime-limited $\eta = 5/4$ target in the unscreened sub-Compton metric-$f(R)$ regime for FST, vs. torsion-dependent corrections for CCGG).

Remark 5.26 (Observational forecasts for $\eta = 5/4$). The gravitational slip parameter $\eta = \Psi/\Phi_N$ provides a sharp discriminant between the regime-limited FST target ($\eta = 5/4$ in the quasi-static, linear, unscreened sub-Compton metric-$f(R)$ regime) and ΛCDM ($\eta = 1$). Current and planned surveys offer the following sensitivities, assuming the relevant modes remain in that unscreened regime:

- *Euclid* (DR1 expected 2027): $\sigma(\eta) \approx 0.03$–$0.05$ from weak lensing + galaxy clustering, potentially sufficient to distinguish $\eta = 5/4$ from $\eta = 1$ if scale cuts and screening systematics are controlled.
- *DESI* (DR1 2025): $\sigma(\eta) \approx 0.08$ from BAO + RSD, marginal ($\sim 3\sigma$) discrimination.
- *LISA* (2030s): Gravitational wave standard sirens provide an independent H_0 measurement with $\sigma(H_0)/H_0 \sim 1\%$, constraining the Hubble tension and indirectly testing η.

A detection of $\eta > 1.1$ at $> 3\sigma$ on unscreened sub-Compton modes would constitute strong evidence for modified gravity; $\eta \approx 1.25$ would be a targeted consistency signal for the FST/$f(R)$ framework, not by itself a proof of the full RFEP mechanism.

Remark 5.27 (Error budget and systematic uncertainties). The theoretical prediction $\eta = 5/4$ is subject to several systematic uncertainties:

1. *Standard Model spectrum:* The particle content determines the one-loop running of Λ. BSM physics (e.g., additional scalars) would shift η by $O(n_{\mathrm{BSM}}/n_{\mathrm{SM}}) \sim 1\%$.
2. *Sub-Compton and screening regime:* For $k < am_s$, or in screened high-density regions where the scalar degree of freedom decouples, the target $\eta = 5/4$ breaks down and $\eta \to 1$ (GR recovery). The transition scale $k_{\mathrm{cross}} = am_s$ depends on γ and on the ambient-density dependence of m_s.
3. *Renormalisation scheme:* The precise value of γ_0 is scheme-dependent at the $O(10\%)$ level. The coefficient ratio producing $\eta = 5/4$ is stable only within the linearised metric-$f(R)$ sub-Compton approximation, not across arbitrary nonlinear screening regimes.

Remark 5.28 (Thermodynamic holography). The UV divergence of the cosmological constant ($\Lambda_0 \sim M_{\mathrm{Pl}}^4$) can be reinterpreted holographically: the M_{Pl}^4 contribution counts bulk degrees of freedom, while the physical $\Lambda_{\mathrm{eff}} \sim H_0^2$ counts boundary degrees of freedom on the de Sitter horizon (Bekenstein–Hawking entropy $S = A/(4G) \sim H_0^{-2}/G$). The FST functional $\Phi[\rho]$ provides the bulk-to-boundary map: minimising Φ over bulk configurations projects onto the boundary-dominated vacuum, absorbing the UV divergence into the renormalisation of ρ_0.

Remark 5.29 (Asymptotic safety and the non-Gaussian UV fixed point)*.* In the asymptotic safety programme (Reuter, Wetterich), the gravitational RG flow possesses a non-Gaussian UV fixed point with $\Lambda^*/k^2 = \text{const}$ and $G^* k^2 = \text{const}$. The IR flow from this fixed point determines Λ_{eff} at $k = H_0$. The FST penalty term $V_{\text{grav}} \sim G\rho^2/k^2$ can be identified with the running cosmological constant at scale k: $\Lambda(k) = 8\pi G(k)\rho^*(k)$. If asymptotic safety holds, the FST functional inherits a natural UV completion, and the large value $\gamma_0 \sim 10^{60}$ is explained by the RG running from the Planck scale to H_0. For a comprehensive review of the asymptotic safety programme and its compatibility with Standard Model matter content, see [11].

Remark 5.30 (Vainshtein screening via Galileon self-interactions)*.* The scalaron field ϕ in the Pöschl–Teller saturation model can be endowed with Galileon-type kinetic self-interactions $\mathcal{L}_{\text{Gal}} = (\partial\phi)^2 \Box\phi/\Lambda_3^3$, where $\Lambda_3 = (M_{\text{Pl}} H_0^2)^{1/3}$ is the Vainshtein scale. Inside the Vainshtein radius $r_V = (M/(4\pi M_{\text{Pl}} \Lambda_3^3))^{1/3}$ of a massive body M, the kinetic terms dominate and the scalaron decouples from matter, recovering GR. This provides a complementary screening mechanism to the chameleon effect (which relies on $m_s(R)$ becoming large), and may be necessary if the Pöschl–Teller potential alone does not suppress the fifth force sufficiently at solar-system scales.

Remark 5.31 (Topological screening via Aharonov–Bohm phases)*.* **Speculative (requires complex scalar field).** For a real scalar field $\phi : M \to \mathbb{R}$, the fundamental group $\pi_1(\mathbb{R}) = 0$ implies that the winding number $\oint \nabla\phi \cdot d\ell/(2\pi)$ vanishes identically by Stokes' theorem for smooth configurations. Non-trivial winding numbers require either a complex scalar field ($\pi_1(U(1)) = \mathbb{Z}$) or a field with singularities. If the dark-energy sector were extended to include a complex scalar field, the resulting winding numbers could provide an Aharonov–Bohm-type screening mechanism independent of the scalaron mass. This possibility remains speculative and requires a significant extension of the model.

Remark 5.32 (Topological RG matching and geometric phase transitions)*.* Two speculative extensions of the RG framework:

1. *Topological RG matching:* The quartic UV divergence $\Lambda_0 \sim M_{\text{Pl}}^4$ may be a topological invariant (analogous to the Atiyah–Singer index), counting the number of fermionic zero modes in the gravitational background. If so, the UV-IR matching is not a dynamical running but a topological identity, and Λ_{eff} is determined by the topology of the de Sitter manifold.

2. *Geometric phase transitions:* The transition from the cosmological regime ($R \sim H_0^2$) to the solar-system regime ($R \sim R_\odot \gg H_0^2$) can be viewed as a spontaneous symmetry breaking in the FST functional: the cosmological de Sitter vacuum breaks the $\gamma(R)$-symmetry, and the solar-system vacuum is a "Higgs phase" with massive scalaron ($m_s \gg H_0$).

Remark 5.33 (Pattern B (Flow Selection) Master Stability – cross-problem structure)*.* This paper instantiates Pattern B (Flow Selection) from the unified framework [20]: strict convexity of the renormalised free energy F, combined with a neutral leading resolvent branch and second-order dominance, implies a spectral gap $\Delta > 0$ for the associated transfer operator. Pattern B shares the variational structure of Pattern A but replaces the static minimiser by an RG flow to a fixed point. The specific instantiation in the present context is: $\Delta =$ curvature of $\Phi[\rho]$ at the de Sitter vacuum. Dark Energy thus instantiates Pattern B, which inherits Pattern A's positivity architecture while selecting the vacuum via a dynamical flow rather than a static energy minimum.

AI Disclosure

In the preparation of this manuscript, AI-assisted tools (Claude, Anthropic) were used in a supporting capacity, particularly for literature research, text structuring, and formulation assistance. The conceptual design, scientific argumentation, and responsibility for the entire content rest solely with the author.

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